

SUPPORTING INFORMATION

Appendix S1 – Ecopath model parameter derivation methodology

The Ecopath with Ecosim approach

We modeled the Warm Pool ecosystem using Ecopath with Ecosim software (www.ecopath.org). Ecopath trophic models describe the static state of energy flows in a food web that balances a group's net production with all sources of mortality and migration (Polovina 1984). The software has a dynamic capability, called *Ecosim*, that allows forecasting of ecosystem responses to specific perturbations through time (Walters *et al.* 1997, Pauly *et al.* 2000). *Ecosim* re-expresses the static *Ecopath* model in a dynamic form, whereby the dynamics and sensitivity of the model is largely controlled by the predator consumption rates and the proportion of the prey that exist in a vulnerable state. Detailed description and discussion of Ecopath and Ecosim can be found in Walters *et al.* (1997), Christensen and Pauly (1992) and Christensen and Walters (2004).

The Ecopath master equation (Equation 1) balances a functional group's net production with all sources of mortality, migration, or change for that group, such that the net production of a functional group equals the sum of (1) the total mass (or energy) removed by predators and fisheries, (2) the net biomass accumulation of the group, (3) the net migration of the group's biomass, and (4) the mass flowing to detritus.

$$B_i \bullet (P/B)_i = \sum_{j=1}^n B_j \bullet (Q/B)_j \bullet DC_{ji} + Y_i + E_i + BA_i + B_i \bullet (P/B)_i \bullet (1 - EE_i) \quad (1)$$

B_i and B_j is the standing biomass (expressed as t km⁻²) of prey (i) and predators (j), respectively, averaged across the modelled region. These parameters were derived using data

from stock assessment models, the SEAPODYM spatial ecosystem and population dynamics model (Lehodey *et al.* 2008), quantitative sampling from scientific surveys, the literature, or if unknown, left to be estimated by Ecopath. The data sources and their preparation for estimating each parameter for each functional group in the model are described in Table S1. Quantitative biomass estimates of intermediate to lower trophic level groups is often a major weakness of pelagic ecosystem models (see Olson and Watters 2003), largely due to the difficulty of quantitatively sampling the standing biomass. Therefore, we used estimates of forage, and phytoplankton biomass from a 2012 version of a SEAPODYM model characterised for skipjack (Senina *et al.* 2008).

P/B_i is the production/biomass ratio (year^{-1}), equivalent to natural mortality (M) for an unfished population, total mortality (Z) for a fished population (Allen 1971), or the inverse of maximum age (ω). If no information was available for natural mortality, which is common, an estimate was made using up to three empirical equations (Pauly 1980, Hoenig 1983, Jensen 1996) depending on the biological data available for a particular group.

Fishing mortality (F) data was available from stock assessments of the primary target species, but was not available for the majority of bycatch species. Therefore, conservative estimates of F were made by assuming $F = 0.8(M)$, which can be a reasonable proxy for fishing mortality at, or below, MSY (Gabriel and Mace 1999). In these cases, initial P/B values were allowed to be decreased if required during the balancing process since they were likely to be overestimated.

EE_i is the ecotrophic efficiency; the fraction of the total production of a group utilized in the system. For most forage groups (e.g. sardines) that are likely to be consumed by a large number of predators there is likely to be only a small proportion of the biomass that is not utilised by the system. Values of around 0.95 are reasonable in such cases (Christensen *et al.* 2009). In contrast, for large predators having few, if any, predators (e.g. large sharks), or very

small, abundant and highly productive organisms (e.g. phytoplankton and zooplankton), only a small percentage of the biomass is likely to be consumed equating to EE values of around 0.1-0.3.

Q/B_j is the annual food consumption rate per unit biomass of j . This is the equivalent of annual ration, or more widely available, the daily ration (in % body weight day⁻¹) multiplied by 365 days. This parameter was either derived from the literature (e.g. Olson and Galvan-Magana 2002, Griffiths *et al.* 2009a), or estimated from von Bertalanffy parameters using the empirical equation of Palomares and Pauly (1998).

DC_{ji} is the proportion of prey i in the average annual diet of predator j , represented in terms of percent wet weight. Diet data were primarily derived from stomach content analyses undertaken by the SPC Oceanic Fisheries Programme (Ecosystem Monitoring and Analysis Section) from samples collected by the observer programmes of the region. These data were supplemented, where necessary, by published and unpublished dietary data collected in studies undertaken in Australia and Papua New Guinea by the CSIRO (e.g. Griffiths *et al.* 2009a, Kloser *et al.* 2009). In the absence of dietary information for cetaceans, we used relevant data from Pauly *et al.* (1998b). Together, these data were used to develop the model's initial diet matrix (Table S2), which was modified during the balancing process to produce the model's final diet matrix (Table S3). The main difference between these two matrices is the lack of information on the predation of high trophic level species, particularly billfish and sharks, in the initial matrix. This is due to the difficulty in documenting such predation events from stomach content analysis, given the likely low frequency of these types of predation events (e.g. sharks consuming swordfish). These predators require predation mortality to be exerted upon them in the model—as they would in the real world—to prevent their populations from unrealistically increasing perpetually. Therefore, we used expert opinion and anecdotal

accounts of predation events involving these species to justify the creation of predatory links in the final diet matrix.

Y_i is the total fisheries catch of a group (expressed as t km^{-2}) averaged across the model region (i.e. $Y = F*B$), and includes both retained and discarded catch. These data were derived from logbook and scientific observer data held by the SPC (e.g. Williams and Terawasi 2015) (Table S4).

E_i is the net migration of i (emigration less immigration) over the course of a year, which was assumed to be zero (i.e. emigration = immigration); and

BA_i is the biomass accumulation rate of i ($\text{t km}^{-2} \text{ year}^{-1}$). This is a production term that can be calculated if the biomass at the end of the year and the beginning of the following year is known. Since Ecopath is generally used as a trophic ‘snap shot’ in time, BA_i is often assumed to be zero (i.e. no biomass accumulation), as it was in the current model, and biomass time series data used in Ecosim (see below) to force dynamic changes in the base Ecopath model.

The Ecopath model is a system with as many such linear equations as there are functional groups. The energy balance (conservation of matter) within each functional group is ensured with the second master equation (Equation 2):

$$\text{Consumption} = \text{production} + \text{respiration} + \text{unassimilated food} \quad (2)$$

Ecosim

An extension of the Ecopath model is the dynamic simulation routines of Ecosim (Walters *et al.* 1997, Walters *et al.* 2000). In Ecosim, information in the static Ecopath model is re-expressed in a dynamic form (Equation 3).

$$\frac{dB_i}{dt} = g_i \sum_j Q_{ji} - \sum_j Q_{ji} + I_i - (MO_i + F_i + e_i) \times B_i \quad (3)$$

$\frac{dB_i}{dt}$ is the change in the biomass of group i (B_i) over time, g_i is the efficiency of the conversion of food into growth, Q_{ji} is the rate of consumption by predators j of prey group i , I_i is the immigration rate, MO_i is the non-predation natural mortality rate, F_i is the fishing mortality rate, and e_i is the emigration rate.

The dynamics and sensitivity of Ecosim models is largely controlled by the consumption rates (Q_{ji}) (Equation 3), which are limited by the proportion of a given predator group's prey that exist in a vulnerable state. Prey vulnerability is controlled within the expression of consumption rate by a user-specified (or calculated) transfer rate of prey movement between vulnerable and invulnerable pools (v_{ij} and v'_{ij}), thus representing the universal community stabiliser of prey refugia. The consumption rate (Q_{ji}) expressed in Equation 4 includes the prey vulnerability parameters.

$$Q_{ji} = \frac{v_{ij} \times a_{ij} \times B_i \times B_j \times T_i \times T_j \times S_{ij} \times \frac{M_{ij}}{D_j}}{v_{ij} + v'_{ij} \times T_i \times M_{ij} + a_{ij} \times M_{ij} \times B_j \times S_{ij} \times \frac{T_j}{D_j}} \quad (4)$$

a_{ij} is the rate of effective search for prey i by predator j , T_i and T_j are the relative feeding times of prey i and predator j , S_{ij} is the user-defined seasonal and long-term environmental forcing effects, M_{ij} is the non-trophic mediation forcing effects, and D_j represents the effects of prey handling time by predators, which further limits consumption rates.

Parameters v_{ij} and v'_{ij} represent prey vulnerabilities—or the rate of exchange of biomass between two prey behavioural states—a state in which all predators have full access to prey

and a state in which prey have full refuge from predators. Prey use refugia in real ecosystems and so not all prey biomass is vulnerable to predation at any given time, and predator-prey relationships are limited by behavioural and physical mechanisms. Ecosim is designed to allow the specification of the type of trophic control (Lotka-Volterra type vs. donor control) that mediates any interaction in the food web. Maximum consumption rates are hypothesised, and thus the rate of exchange of biomass (v_{ij}) that a predator normally exerts. For high prey vulnerability (v_{ij}) the functional relationship approximates a mass-action flow, or Lotka-Volterra type interaction implying a strong ‘top-down’ effect. For low prey vulnerabilities the functional relationship approaches a donor-controlled (bottom-up) flow rate so v_{ij} is the maximum possible instantaneous mortality rate that j can cause on i (Walters *et al.* 1997).

Prey vulnerabilities can be specified by adjusting the proportion of prey in vulnerable and invulnerable states (pools) via adjustment of the v values, which represents Lotka-Volterra (top-down) type control and pure donor control. In the real world, this mixture of trophic control is mediated by temporal or spatial refugia, or by the relative primacy of physical and biotic forces in regulating communities, i.e. predator-prey interactions. The vulnerability value for each predator-prey interaction was estimated by fitting the Ecosim model to time series of biomass, fishing mortality, and catch for functional groups where these data were available (see Appendix S3 “*Calibrating the Ecosim model*”).

Table S1. Description of data and processes undertaken to derive estimates of biomass, the production/biomass ratio (P/B), the consumption biomass ratio (Q/B), ecotrophic efficiency (EE), fishery landings and discards for each functional group in the Ecopath model of the Warm Pool.

Functional group	Data type	Age classes	Source	Data included in the Ecopath model
Seabirds (<i>Phoebastria nigripes</i> , <i>Phoebastria immutabilis</i> , <i>Diomedea</i> spp.)	Biomass	All	Olson and Watters (2003)	In the absence of biomass estimates for the Warm Pool, used estimates of the "pursuit birds" group in the eastern tropical Pacific Ocean from Olson & Watters (2003) of 600 t per million km ² .
	P/B	All	Tuck <i>et al.</i> (2001), (2015), Rivalan <i>et al.</i> (2010)	Used the average adult natural mortality estimate used in population models for wandering albatross (<i>Diomedea exulans</i>) ($M = 0.02 \text{ yr}^{-1}$) and the Amsterdam Albatross (<i>Diomedea amsterdamensis</i>) ($M = 0.029 \text{ yr}^{-1}$). In the absence of estimates of fishing mortality in the Warm Pool, a precautionary estimate of fishing mortality used assuming birds incurred a mortality in the vicinity of MSY estimated as $F = 0.8M = 0.0196 \text{ yr}^{-1}$. Therefore, initial $P/B = 0.0245 + 0.0196 = 0.0441 \text{ yr}^{-1}$.
	Q/B	All	Catry <i>et al.</i> (2004)	Average daily consumption by <i>Diomedea chrysostoma</i> was estimated at 750g d ⁻¹ for birds averaging 3.65kg, equating to a daily ration of 20.55% body weight (BW) day ⁻¹ and an annual ration (Q/B) of 74.99 times the population biomass yr ⁻¹ .
	Landings	All	N/A	Not permitted to be landed in Warm Pool fisheries.
	Discards	All	Molony (2005), Rivalan <i>et al.</i> (2010)	Estimate derived from expanded observer data for the longline and purse seine fisheries in the WCPO for 1990-2004 (Molony, 2005). Longline annual catch estimated at 1593 birds with a 100% mortality rate. Purse seine annual catch estimated at only 20 birds per year, so this did not warrant inclusion. The Warm Pool modelled region is approximately 50% of the WCPO area assessed for bycatch, so catch estimates halved. Using a mean weight of 7 kg for the most commonly caught species (Albatross, <i>Diomedea</i> spp.), the 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²) was 0.000000461 km ² .
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Gould <i>et al.</i> (1997), Harrison <i>et al.</i> (1983), Cherel and Klages (1998)	In the absence of dietary studies of seabirds in the Warm Pool, used diet composition (by % weight) of six species of Diomeduids in the North Pacific Ocean (Hawaii), which was dominated by epipelagic squids (20-67%), fishes (9-50%), and crustaceans (5-40%) to a lesser extent migratory mesopelagic fishes, crustaceans and fishery discards. Many of these species migrate across the entire Pacific Ocean, so these diets are likely to be representative in the Warm Pool at the functional group level.
Cetaceans (<i>Stenella longirostris</i> , <i>Tursiops truncatus</i> , <i>Delphinus delphis</i> , <i>Grampus griseus</i> , <i>Globicephala macrorhynchus</i>)	Biomass	All	Griffiths <i>et al.</i> (2010)	In the absence of biomass estimates for cetaceans for the Warm Pool, initially used estimate for cetaceans in a model of the pelagic ecosystem off eastern Australia (0.013 t km ⁻²). However, given the confidence in P/B and Q/B estimates, this biomass was far too high, producing an EE of only 0.002, which should be around 0.1-0.3 for apex predators having few predators (Christensen <i>et al.</i> 2009). The biomass of cetaceans would also be expected to be lower in the Warm Pool due to its oligotrophic waters. Therefore, during the balancing process biomass was iteratively reduced until EE was around 0.1.
	P/B	All	Danil and Chivers (2007)	In absence of P/B estimates for the Warm Pool, produced estimate from an age and growth study of <i>Delphinus delphis</i> in the eastern tropical Pacific Ocean using von Bertalanffy parameters $L_{\infty} = 197.2$, $K = 0.687 \text{ yr}^{-1}$ and the natural mortality empirical equation of Pauly (1980) using a water temperature of 29° C to produce $M = 0.8385 \text{ yr}^{-1}$. Since this stock is not fished and rarely caught as bycatch, so F assumed to be zero. Thus $P/B = 0.8385 \text{ yr}^{-1}$.
	Q/B	All	Sergeant (1969)	Daily ration of adult <i>Tursiops truncatus</i> in captivity was estimated to be 4-6% BW day ⁻¹ . Used the lower value (4%) as cetaceans most likely encounter prey less frequently in the wild. Thus $Q/B = 14.60 \text{ yr}^{-1}$.
	Landings	All	N/A	Not permitted to be landed in Warm Pool fisheries.
	Discards	All	Molony (2005)	Estimate derived from expanded observer data for the longline and purse seine fisheries in the WCPO for 1990-2004. Longline annual catch estimated at 2200 animals with a post release mortality of 30%. Purse seine annual catch estimated at 3500 with a mortality rate of 10%, with 85% (or 2975 animals) being caught in PSA sets. The Warm Pool modelled region is approximately 50% of the WCPO area assessed for bycatch, so the total annual catch by fishery was: longline – 1100, PSA – 1488, PSU – 262. Using a mean weight of 100 kg for the most commonly-caught species (Common dolphin, <i>Delphinus delphis</i>), the 2005

				estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²) was 0.0000091 km ⁻² , 0.00001231 km ⁻² , 0.000002172 km ⁻² , respectively.
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Dolar <i>et al.</i> (2003), Perrin <i>et al.</i> (1973), Pitman and Stinchcomb (2002),	Little quantitative dietary data is available for cetaceans in the Warm Pool, except for <i>Stenella longirostris</i> in the Sulu Sea, Philippines (Dolar <i>et al.</i> 2003). This data was supplemented by with diet composition (by % weight or volume) of <i>Stenella</i> spp. in the eastern tropical Pacific Ocean. Most of these are present in the western Pacific Ocean, so their eastern Pacific diets are likely to be representative in the Warm Pool at the functional group level. Establishment of additional predation linkages with some functional groups (e.g. dolphinfish) was necessary based on qualitative or anecdotal evidence since very little supporting quantitative information is available due to few dietary studies on cetaceans and the low incidence of predation by these larger predators.
Turtles (<i>Caretta caretta</i> , <i>Chelonia mydas</i> , <i>Lepidochelys olivacea</i> , <i>Eretmochelys imbricate</i>)	Biomass	All	Griffiths <i>et al.</i> (2010), Okey (2006)	Used 0.0033 t km ⁻² for turtles from tropical northern and eastern Australia. Combines estimates for <i>Chelonia mydas</i> (0.0015 t km ⁻²) and <i>Dermochelys coriacea</i> (0.0018 t km ⁻²).
	P/B	All	Limpus (1992), Wabnitz and Pauly (2008)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) for <i>Eretmochelys imbricata</i> from tropical Australian waters of L_{∞} = 100 cm CL and K = 0.048 yr ⁻¹ (Limpus 1992) at 29° C water temperature, which gave M = 0.177 yr ⁻¹ . <i>Caretta caretta</i> from the Great Barrier Reef, Australia has an L_{∞} = 105 cm CL and K = 0.06 yr ⁻¹ (Wabnitz and Pauly 2008) at 29° C water temperature gave M = 0.202 yr ⁻¹ . An average was used (M = 0.190 yr ⁻¹) given the presumption there should be no fishing mortality on turtles.
	Q/B	All	Njoman <i>et al.</i> (1982)	Estimate of 1.01% BW yr ⁻¹ was experimentally derived for <i>Caretta caretta</i> at 20.5° C water temperature by Njoman <i>et al.</i> (1982). Daily ration in warmer waters of the Warm Pool (>29° C) is likely to be considerably higher, so 2% BW yr ⁻¹ was used, thus Q/B = 7.3 yr ⁻¹ .
	Landings	All	N/A	Not permitted to be landed in Warm Pool fisheries.
	Discards	All	Unpublished SPC scientific observer data (Valerie Allain - SPC)	The number of turtles discarded dead or with unknown status in the longline fishery was estimated based on longline observer data in the Warm Pool area for the period 2000-2010 and raised to the fleet according to the effort coverage. No estimate provided for the purse seine and pole-and-line fisheries.
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Bjorndal (1997), Heithaus <i>et al.</i> (2008), Boyle and Limpus (2008)	Averaged diet proportions across three main species <i>Caretta caretta</i> , <i>Chelonia mydas</i> , and <i>Eretmochelys imbricate</i> from studies conducted in the western and northern Pacific Ocean, primarily Australia and Hawaii, but probably representative of the diets of turtles in the Warm Pool at the functional group level. Predation on turtles however, is not well documented quantitatively for adults or juveniles, but reliable qualitative studies (Heithaus <i>et al.</i> , 2008, Boyle and Limpus, 2008) justify the establishment of trophic linkages to relevant predators in the model.
Small swordfish (<i>Xiphias gladius</i>)	Biomass	1-2 yrs	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	Estimate derived from the 2013 Multifan-CL (MFCL) age structured stock assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) for age classes 1-2 years in Area 1 in 2005. Biomass estimate of 2760.97 t was divided by ocean surface area of Area 1 (10,339,000 km ²) to produce an initial estimate of 0.000267 t km ⁻² .
	P/B	1-2 yrs	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	The 2013 MFCL age structured stock assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) estimates natural mortality (M) of age class 1- 2 years as 0.23875 yr ⁻¹ . Fishing mortality (F) for small fish in 2005 in Area 1 was 0.0388 yr ⁻¹ . Therefore, $M + F = Z \approx$ P/B was 0.2776 yr ⁻¹ . P/B was too low to derive a reliable P/Q estimate, assuming daily ration was reliable. P/B would be expected to be higher in smaller fish than larger fish, so value allowed to increase slightly.
	Q/B	1-2 yrs	Stillwell and Kohler (1985)	Daily ration of swordfish in the Atlantic 0.94% to 1.6% body weight (BW) day ⁻¹ (Stillwell and Kohler 1985), so 1.3% BW day ⁻¹ (Q/B = 4.745 yr ⁻¹) was used in order to produce P/Q estimates of around 0.1 yr ⁻¹ .
	Landings	1-2 yrs	Unpublished SPC scientific observer data (Tim Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	1-2 yrs	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine

				observer data in the WCPO during the period 1995-2005. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	1-2 yrs	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Dambacher <i>et al.</i> (2010), Young <i>et al.</i> (2006), (2010)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data in the Warm Pool and central Pacific Ocean (Allain, 2010, Dambacher <i>et al.</i> 2010). Diet matrix supplemented with trophic connections to less frequently encountered prey in other functional groups based on studies in the adjacent tropical Coral Sea (Young <i>et al.</i> 2006, 2010) and anecdotal accounts (e.g. observed predation by sharks or cetaceans).
Large swordfish (<i>Xiphias gladius</i>)	Biomass	3-20 yrs	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	Estimate derived from the 2013 MFCL age structured stock assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) for age classes 3-20 years in Area 1 in 2005. Biomass estimate of 17922.36 t was divided by the ocean surface area of Area 1 (10,339,000 km ²) to produce an estimate of 0.00173 t km ⁻²
	P/B	3-20 yrs	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	The 2013 MFCL age structured stock assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) estimates natural mortality (<i>M</i>) of age class 3-20 years as 0.154 yr ⁻¹ . Fishing mortality (<i>F</i>) in Area 1 of the 2013 MFCL assessment for large fish in 2005 was 0.140 yr ⁻¹ . Therefore, $M + F = Z \approx P/B$ was 0.293 yr ⁻¹ .
	Q/B	3-20 yrs	Stillwell and Kohler (1985)	Daily ration of swordfish in the Atlantic 0.94% to 1.6% body weight (BW) day ⁻¹ (Stillwell and Kohler 1985), so 0.94% BW day ⁻¹ day (Q/B = 3.431 yr ⁻¹) was used in order to produce P/Q estimates of around 0.1 yr ⁻¹ .
	Landings	3-20 yrs	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	3-20 yrs	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	3-20 yrs	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Dambacher <i>et al.</i> (2010), Young <i>et al.</i> (2006), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data in the Warm Pool and central Pacific Ocean (Allain, 2010, Dambacher <i>et al.</i> 2010). Diet matrix supplemented with trophic connections to less frequently encountered prey in other functional groups based on studies in the adjacent tropical Coral Sea (Young <i>et al.</i> 2006, 2010) and anecdotal accounts (e.g. observed predation by sharks or cetaceans).
Blue marlin (<i>Makaira nigricans</i>)	Biomass	All	Kleiber <i>et al.</i> (2003), Simon Hoyle (SPC)	Estimate derived from the SW area combination of two blocks in a MFCL Pacific-wide stock assessment for blue marlin (Kleiber <i>et al.</i> 2003) for all age classes combined. The 2005 biomass estimate of 11148.26 t was divided by the ocean surface area of the assessment area (41,140,000 km ²) to produce an estimate of 0.000270984 t km ⁻² .
	P/B	All	Sun <i>et al.</i> (2013), Kleiber <i>et al.</i> (2003), Simon Hoyle (SPC)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) using $L_{\infty} = 236.1$ cm LJFL and $K = 0.145$ yr ⁻¹ from Sun <i>et al.</i> (2013) and using a water temperature of 29° C to produce $M = 0.274$ yr ⁻¹ . Used $F = 0.034$ yr ⁻¹ for the SW area combination of 2 blocks in the MFCL age structured stock assessment by Kleiber <i>et al.</i> (2003). Therefore, $M + F = Z \approx P/B$ was 0.4291 yr ⁻¹ .
	Q/B	All	Abitia-Cárdenas <i>et al.</i> (2011)	No value available for <i>Makaira nigricans</i> in the Warm Pool, so daily ration assumed to be similar to <i>Kajikia audax</i> in tropical Mexican waters, estimated to be 3.6% BW day ⁻¹ (Q/B = 13.14 yr ⁻¹) (Abitia-Cárdenas <i>et al.</i> 2011). However, <i>M. nigricans</i> grows significantly larger than <i>K. audax</i> and therefore is likely to have a lower daily ration. Thus, it was assumed daily ration was 2% BW day ⁻¹ (Q/B = 7.30 yr ⁻¹).
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine

				observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Shimose <i>et al.</i> (2006)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data for <i>Makaira nigricans</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with diet data from a comprehensive quantitative study of <i>Makaira nigricans</i> in the adjacent tropical East China Sea and Philippine Sea (Shimose <i>et al.</i> 2006).
Striped marlin (<i>Kajikia audax</i>)	Biomass	All	Davies <i>et al.</i> (2012), Simon Hoyle (SPC)	Estimate derived from the South Pacific area of the 2012 MFCL stock assessment for the southwestern Pacific Ocean (Davies <i>et al.</i> 2012). The 2005 biomass estimate of 5492.69 t for all age classes combined was divided by the water surface area of 42,973,000 km ² of the assessment area. The initial value used was 0.000127817 t km ⁻² . Reduced value slightly to 6.2 x 10 ⁻⁵ t km ⁻² in order for Ecopath's <i>F</i> value to match MFCL <i>F</i> value in 2005.
	P/B	All	Kopf <i>et al.</i> (2005) Davies <i>et al.</i> (2012), Simon Hoyle (SPC)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) using $L_{\infty} = 233.6$ cm LJFL and $K = 0.440$ yr ⁻¹ from Kopf <i>et al.</i> (2005) study in the SW Pacific with water temperature of 29° C to produce $M = 0.5975$ yr ⁻¹ . $F = 0.304$ yr ⁻¹ in the South Pacific area of the 2012 MFCL stock assessment for the southwestern Pacific Ocean (Davies <i>et al.</i> 2012). Therefore, $M + F = Z \approx P/B$ was 0.902 yr ⁻¹ .
	Q/B	All	Abitia-Cárdenas <i>et al.</i> (2011)	No value available specifically for <i>Kajikia audax</i> in the Warm Pool, so used daily ration of <i>Kajikia audax</i> studied in tropical Mexican waters estimated to be 3.6% BW day ⁻¹ (Q/B = 13.14 yr ⁻¹) (Abitia-Cárdenas <i>et al.</i> 2011).
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Shimose <i>et al.</i> (2010), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data for <i>Kajikia audax</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>Kajikia audax</i> in the adjacent tropical East China Sea and Philippine Sea (Shimose <i>et al.</i> 2010), and Coral Sea (Young <i>et al.</i> 2010).
Other billfish (<i>Istiompax indica</i> <i>Istiophorus platypterus</i> <i>Tetrapterus</i> spp.)	Biomass	All	See “Striped marlin” and “Blue marlin”	Assumed that sailfish and spearfish were as least as abundant as blue marlin and that the other species in the group were probably as abundant as striped marlin, which is reasonably low in the Warm Pool. Therefore, biomass of these groups summed to give an initial estimate of 0.000399 t km ⁻² . Biomass decreased to 0.00022 t km ⁻² in order for EE to be more realistic being lower than that of sharks, which are marlin predators.
	P/B	All	See “Striped marlin”	Assumed small billfish like sailfish and other <i>Tetrapterus</i> species have reasonably high P/B, so P/B assumed to be the same as striped marlin = 0.902 yr ⁻¹ .
	Q/B	All	Abitia-Cárdenas <i>et al.</i> (2011)	No value available for other billfish (e.g. sailfish and spearfish) in the Warm Pool, so daily ration assumed to be the same as for <i>Kajikia audax</i> in tropical Mexican waters estimated to be 3.6% BW day ⁻¹ (Q/B = 13.14 yr ⁻¹) (Abitia-Cárdenas <i>et al.</i> 2011).
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).

	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Shimose <i>et al.</i> (2010), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents for billfish caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>Istiophorus platypterus</i> and <i>Tetrapturus angustirostris</i> in the central Pacific Ocean (Shimose <i>et al.</i> 2010), and Coral Sea (Young <i>et al.</i> 2010).
Mako shark (<i>Isurus oxyrinchus</i>)	Biomass	All	Rice and Harley (2012)	In the absence of species specific biomass estimates, it was assumed mako sharks would have a similar biomass as Oceanic White Tip, which was estimated using a stock assessment model. Therefore, the value of 0.000294 t km ⁻² was used.
	P/B	All	Ribot-Carballal <i>et al.</i> (2005)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) for <i>Isurus oxyrinchus</i> from Baja, Mexico (Ribot-Carballal <i>et al.</i> 2005) of L_{∞} = 411 cm FL and K = 0.05 yr ⁻¹ at 29° C water temperature which gave M = 0.123. To calculate P/B of this fished stock with no available value of F , it was assumed F = 0.8 M if stock is fished at or below MSY = 0.221 yr ⁻¹ . Therefore, the initial P/B = 0.344 yr ⁻¹ . This value was required to be increased to 0.577 yr ⁻¹ in order for P/Q to approach 0.1 for high order predators.
	Q/B	All	Wood <i>et al.</i> (2009)	Daily ration of mako shark in the northwest Atlantic Ocean estimated by Wood <i>et al.</i> (2009) as 4.6% BW day ⁻¹ (Q/B = 16.79 yr ⁻¹). This value was reduced to 7.30 yr ⁻¹ in order to produce P/Q estimates of around 0.1 yr ⁻¹ .
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data for <i>Isurus oxyrinchus</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from a comprehensive quantitative dietary study of <i>I. oxyrinchus</i> in the adjacent tropical Coral Sea (Young <i>et al.</i> 2010).
Blue shark (<i>Prionace glauca</i>)	Biomass	All	Kleiber <i>et al.</i> (2009)	Estimate derived from Region 3 for all age classes of the North Pacific blue shark MFCL stock assessment (Kleiber <i>et al.</i> 2009). Biomass of 0.01038 t km ⁻² in 2002 was the most recent estimate available. However, biomass in the Warm Pool this is likely to be much lower in 2005 given the increase in catch and fishing mortality and because blue sharks prefer cooler waters.
	P/B	All	Lessa (2004)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) for <i>Prionace glauca</i> from NE Brazil (Lessa 2004) of L_{∞} = 352.1 cm FL and K = 0.157 yr ⁻¹ at 29° C water temperature which gave M = 0.272. To calculate P/B of this fished stock with no available value of F for 2005, it was assumed F = 0.8 M if stock is fished at or below MSY = 0.218 yr ⁻¹ . Therefore, the initial P/B = 0.490 yr ⁻¹ . This value was required to be decreased to 0.440 yr ⁻¹ in order for P/Q to approach the expected 0.1 for high order predators.
	Q/B	All	Kohler (1988)	In the absence of daily ration estimates from the Warm Pool or Pacific Ocean, used daily ration estimate of 0.06% BW day ⁻¹ (Q/B = 2.19 yr ⁻¹) for <i>Prionace glauca</i> in the Atlantic Ocean, but increased to 3.0 given the higher water temperatures in the Warm Pool.
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).

	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Young <i>et al.</i> (2010), Kubodera <i>et al.</i> (2006)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data for <i>Prionace glauca</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>P. glauca</i> in the adjacent tropical Coral Sea (Young <i>et al.</i> 2010) and western Pacific Ocean (Kubodera <i>et al.</i> 2006).
Silky shark (<i>Carcharhinus falciformis</i>)	Biomass	All	Rice and Harley (2013)	Biomass estimate derived from the WCPO area bounded by 30°N-30°S, a combination of 2 blocks in a Stock Synthesis stock assessment model for silky shark (Rice and Harley 2013). The 2005 biomass value of 49369.6 t was used, which covered all age classes. Biomass was divided by the ocean surface area of 69,186,000 km ² to provide the initial value of 0.000714 t km ⁻² .
	P/B	All	Oshitani <i>et al.</i> (2003) Rice and Harley (2013)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) using $L_{\infty} = 216.4$, $K = 0.148 \text{ yr}^{-1}$ from Oshitani <i>et al.</i> (2003) in the Pacific Ocean and using a water temperature of 29° C to produce $M = 0.297 \text{ yr}^{-1}$. Used $F = 0.280 \text{ yr}^{-1}$ for the WCPO area bounded by 30°N-30°S, a combination of 2 blocks in a Stock Synthesis stock assessment model for silky shark (Rice and Harley 2013), which covered all age classes. Therefore, $M + F = Z \approx P/B$ was 0.577 yr^{-1} .
	Q/B	All	Oshitani <i>et al.</i> (2003) Palomares and Pauly (1998)	Used von Bertalanffy parameters and the empirical equation of Palomares and Pauly (1998) using $L_{\infty} = 216.4$, $K = 0.148 \text{ yr}^{-1}$ from Oshitani <i>et al.</i> (2003) in the Pacific Ocean and using a water temperature of 29° C to produce a Q/B of 3.5 times the population biomass yr^{-1} .
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Cabrera-Chávez-Costa <i>et al.</i> (2010), Duffy <i>et al.</i> (2015)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data for <i>Carcharhinus falciformis</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010). Little other data available for the region so the data matrix was supplemented with data from comprehensive quantitative dietary studies of <i>C. falciformis</i> in the tropical eastern Pacific Ocean (Cabrera-Chávez-Costa <i>et al.</i> 2010; Duffy <i>et al.</i> 2015).
Oceanic whitetip shark (<i>Carcharhinus longimanus</i>)	Biomass	All	Rice and Harley (2012)	Biomass estimate derived from the WCPO area bounded by 30°N-30°S, a combination of 2 blocks of a Stock Synthesis stock assessment model for White tip shark (Rice and Harley 2012). The 2005 biomass value of 20327.70 t was used, which covered all age classes. Biomass was divided by the ocean surface area of 69,186,000 km ² to provide the initial value of 0.000294 t km ⁻² .
	P/B	All	Rice and Harley (2012)	Used von Bertalanffy parameters for the similar silky shark and the natural mortality empirical equation of Pauly (1980) using $L_{\infty} = 216.4$, $K = 0.148 \text{ yr}^{-1}$ from Oshitani <i>et al.</i> (2003) in the Pacific Ocean and using a water temperature of 29° C to produce $M = 0.297 \text{ yr}^{-1}$. Used $F = 0.395 \text{ yr}^{-1}$ for 2005 for the WCPO area bounded by 30°N-30°S, a combination of 2 blocks of a Stock Synthesis stock assessment model for White tip shark (Rice and Harley 2012), which covered all age classes. Therefore, $M + F = Z \approx P/B$ was 0.692 yr^{-1} .
	Q/B	All	Stillwell and Kohler (1982)	Used daily ration estimates for a similar species in the Atlantic Ocean, <i>Carcharhinus plumbeus</i> , of $0.9\% \text{ BW day}^{-1}$ (Q/B = 3.29 yr^{-1}) (Stillwell and Kohler 1982).
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the

				Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Bonfil <i>et al.</i> (2008)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data for <i>Carcharhinus longimanus</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010). Little other data available for adjacent regions or other oceans, besides qualitative descriptions of prey consumed (Bonfil <i>et al.</i> 2008), which was integrated into the diet matrix where appropriate.
Other sharks (Lamniformes, Carcharhiniformes)	Biomass	All	See “Silky shark” and “White-tip shark”	In the absence of biomass data, it was assumed the various species of primarily carcharhinid sharks, threshers and hammerhead sharks in the group may be equal to the sum of the biomass of the two other carcharhinid sharks in the model (i.e. Silky and Oceanic whitetip). The initial biomasses of 0.0008 t km ⁻² was decreased to 0.0005 t km ⁻² during balancing to bring EE to around 0.3.
	P/B	All	Oshitani <i>et al.</i> (2003)	In the absence of reliable fishery mortality or biological data, it was assumed the various species of primarily carcharhinid sharks, threshers and hammerhead sharks in the group would have a similar life history as the silky shark (see Oshitani <i>et al.</i> 2003), so we used $M = 0.297 \text{ yr}^{-1}$. To calculate P/B of this lightly fished stock with no available value of F for 2005, it was assumed $F = 0.8M$ if stock is fished at or below $MSY = 0.238 \text{ yr}^{-1}$. Therefore, the initial P/B = 0.535 yr ⁻¹ . Knowing this was likely to be an overestimate, this value was incrementally decreased to 0.450 yr ⁻¹ where the P/Q value was 0.1, which is appropriate for high order predators.
	Q/B	All	See “Silky shark”	This group of hammerheads, threshers and carcharhinids is likely to have a similar Q/B to silky and white tip sharks but probably not as high as high performance pelagic lamnid sharks. Therefore, a Q/B of 4.5 yr ⁻¹ was used.
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Heithaus and Dill (2002), Heithaus <i>et al.</i> (2008), (Kubodera <i>et al.</i> 2006)	Used diet proportions (in % weight) from published and unpublished SPC stomach contents data for species of Lamniformes and Carcharhiniformes caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>Lamna ditropis</i> in the adjacent western Pacific Ocean (Kubodera <i>et al.</i> 2006). Establishment of additional predation linkages with some functional groups (e.g. turtles and cetaceans) was necessary based on qualitative or anecdotal evidence since very little supporting quantitative information is available due to few dietary studies on cetaceans and the low incidence of predation by these larger predators (see Heithaus and Dill, 2002, Heithaus <i>et al.</i> 2008).

Small bigeye tuna (<i>Thunnus obesus</i>)	Biomass	1-16 qts	Davies <i>et al.</i> (2011), Simon Hoyle (SPC)	Biomass estimate of small BET (age classes 1-16 quarters) in 2005 was taken from Region 3 in the 2011 MFCL stock assessment (Davies <i>et al.</i> 2011). The biomass was divided by Region 3's ocean surface area of 21,721,000 km ² = 0.00720 t km ⁻² . This group was the leading stanza of the two-stanza group for bigeye tuna in the Ecopath model where entries were required for biomass, total mortality ($Z \approx P/B$, and Q/B). Given F and M were reliable from fishery data and daily ration ($=Q/B/365$) available from a bioenergetics model (Kirby 2009), these parameters were fixed and biomass adjusted during balancing, which resulted in a biomass of 0.00435 t km ⁻² .
	P/B	1-16 qts	Davies <i>et al.</i> (2011), Simon Hoyle (SPC) Farley <i>et al.</i> (2003)	Fishing mortality (F) in Region 3 of the 2011 MFCL stock assessment for western Pacific bigeye tuna (Davies <i>et al.</i> 2011) for small BET (age 1-16 quarters) in 2005 was 0.560 yr ⁻¹ . Natural mortality (M) estimate of 0.4304 yr ⁻¹ derived for small BET in the south-west Pacific Ocean from the empirical equation of Pauly (1980) using the von Bertalanffy growth parameters $L_{\infty} = 169.09$ cm FL, $K = 0.238$ yr ⁻¹ derived from Farley <i>et al.</i> (2003) and using a water temperature of 29° C assuming small fish remain primarily above the thermocline. Therefore, $M + F = Z \approx P/B$ was 0.990 yr ⁻¹ .
	Q/B	1-16 qts	Olson and Boggs (1986)	Used average daily ration estimate of 4.0% BW day ⁻¹ (Q/B of 14.6 times the population biomass yr ⁻¹) that was experimentally determined for small <i>Thunnus albacares</i> (Olson & Boggs, 1986), but decreased to 3.0% BW day ⁻¹ (Q/B = 10.95 yr ⁻¹) given metabolic costs of occupying deeper and cooler waters than <i>Thunnus albacares</i> .
	Landings	1-16 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that all BET caught in the pole-and-line fishery were small BET. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	1-16 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that all BET caught in the pole-and-line fishery were small BET. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	1-16 qts	Ecopath	Estimated by Ecopath.
	Diet	1-16 qts	Allain (2010), Duffy <i>et al.</i> (2017), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for small <i>Thunnus obesus</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>T. obesus</i> in the adjacent tropical Coral Sea (Young <i>et al.</i> 2010).
Large bigeye tuna (<i>Thunnus obesus</i>)	Biomass	17-40 qts	Davies <i>et al.</i> (2011), Simon Hoyle (SPC)	Biomass estimate of large BET (age classes 17-40 quarters) in 2005 was taken from Region 3 in the 2011 MFCL stock assessment (Davies <i>et al.</i> 2011). The biomass was divided by Region 3's ocean surface area of 21,721,000 km ² = 0.00438 t km ⁻² . This group was the dependent stanza of the two-stanza group for bigeye tuna in the Ecopath model whereby entries for the small BET stanza dictated parameter values for biomass, total mortality ($Z \approx P/B$, and Q/B). The model estimated a biomass of 0.00367 t km ⁻² .
	P/B	17-40 qts	Davies <i>et al.</i> (2011), Simon Hoyle (SPC), Farley <i>et al.</i> (2003)	Fishing mortality (F) in Region 3 of the 2011 MFCL stock assessment for western Pacific bigeye tuna (Davies <i>et al.</i> 2011) for large BET (age 17-40 quarters) in 2005 was 0.1384 yr ⁻¹ . Natural mortality (M) estimate of 0.3320 yr ⁻¹ derived for large BET in the south-west Pacific Ocean from the empirical equation of Pauly (1980) using the von Bertalanffy growth parameters $L_{\infty} = 169.09$ cm FL, $K = 0.238$ yr ⁻¹ derived from Farley <i>et al.</i> (2003) and using a water temperature of 16° C assuming large fish spend greatest time below the thermocline Therefore, $M + F = Z \approx P/B$ was 0.470 yr ⁻¹ .
	Q/B	17-40 qts	Ecopath multi-stanza	For "Small bigeye", Used average daily ration estimate of 4.0% BW day ⁻¹ (Q/B of 14.6 times the population biomass yr ⁻¹) that was experimentally determined for small <i>Thunnus albacares</i> (Olson & Boggs, 1986), but decreased to 3.0% BW day ⁻¹ (Q/B =

				10.95 yr ⁻¹) given metabolic costs of occupying deeper and cooler waters than <i>Thunnus albacares</i> . “Small bigeye” was the leading stanza in the multi-stanza model in Ecopath. Ecopath estimates Q/B for this group.
	Landings	17-40 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	17-40 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	17-40 qts	Ecopath	Estimated by Ecopath.
	Diet	17-40 qts	Allain (2010), Duffy <i>et al.</i> (2017), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for large <i>Thunnus obesus</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>T. obesus</i> in the adjacent tropical Coral Sea (Young <i>et al.</i> 2010).
Small yellowfin tuna (<i>Thunnus albacares</i>)	Biomass	1-9 qts	Langley <i>et al.</i> (2011), Simon Hoyle (SPC)	Biomass estimate for small YFT (age classes 1-9 quarters) in 2005 was taken from Region 3 in the 2011 MFCL stock assessment (Langley <i>et al.</i> 2011). The biomass was divided by Region 3’s ocean surface area of 21,721,000 km ² = 0.0011 t km ⁻² . This group was the dependent stanza of the two-stanza group for yellowfin tuna in the Ecopath model whereby entries for the large YFT stanza dictated parameter values for biomass, total mortality ($Z \approx P/B$), and Q/B. The model estimated a biomass of 0.0275 t km ⁻² .
	P/B	1-9 qts	Langley <i>et al.</i> (2011), Simon Hoyle (SPC), Lehodey and Leroy (1999)	Fishing mortality (F) in Region 3 of the 2011 MFCL stock assessment for western Pacific yellowfin tuna (Langley <i>et al.</i> 2011) for large YFT (age 1-9 quarters) in 2005 was 0.49720 yr ⁻¹ . Natural mortality (M) estimate of 0.9056 yr ⁻¹ derived for all age classes combined of YFT in the western Pacific Ocean from the empirical equation of Pauly (1980) using conservative von Bertalanffy growth parameters for males; $L_{\infty} = 146.7$ cm FL, $K = 0.805$ yr ⁻¹ derived from Lehodey and Leroy (1999) and using a mean water temperature of 29° C in the study region. Therefore, $M + F = Z \approx P/B$ was 1.403 yr ⁻¹ .
	Q/B	1-9 qts	Olson and Boggs (1986)	Used average daily ration estimate of 4.0% BW day ⁻¹ (Q/B of 14.6 times the population biomass yr ⁻¹) which was experimentally determined for small <i>Thunnus albacares</i> (Olson & Boggs, 1986).
	Landings	1-9 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that all YFT caught in the pole-and-line fishery was small BET. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	1-9 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that all YFT caught in the pole-and-line fishery was small BET. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	1-9 qts	Ecopath	Estimated by Ecopath.
	Diet	1-9 qts	Allain (2010), Duffy <i>et al.</i> (2017), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for small <i>Thunnus albacares</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>T. albacares</i> in the adjacent tropical Coral Sea (Young <i>et al.</i> 2010).
Large yellowfin tuna	Biomass	10-28 qts	Langley <i>et al.</i> (2011), Simon Hoyle (SPC),	Biomass estimate for large YFT (age 10-28 quarters) in 2005 was taken from Region 3 in the 2011 MFCL stock assessment (Langley <i>et al.</i> 2011). The biomass was divided by Region 3’s ocean surface area of 21,721,000 km ² = 0.0086 t km ⁻² . This group

<i>(Thunnus albacares)</i>				was the leading stanza of the two-stanza group for yellowfin tuna in the Ecopath model where entries were required for biomass, total mortality ($Z \approx P/B$, and Q/B). Given F and M were reliable from fishery data and daily ration ($[Q/B]/365$) available from a bioenergetics model, these parameters were fixed and biomass adjusted during balancing, which resulted in a biomass of $0.00419 \text{ t km}^{-2}$.
	P/B	10-28 qts	Langley <i>et al.</i> (2011), Simon Hoyle (SPC), Lehodey and Leroy (1999)	Fishing mortality (F) in Region 3 of the 2011 MFCL stock assessment for western Pacific yellowfin tuna (Langley <i>et al.</i> 2011) for large YFT (age 10-28 quarters) in 2005 was 0.765 yr^{-1} . Natural mortality (M) estimate of 0.906 yr^{-1} derived for all age classes combined of YFT in the western Pacific Ocean from the empirical equation of Pauly (1980) using conservative von Bertalanffy growth parameters for males; $L_{\infty} = 146.7 \text{ cm FL}$, $K = 0.805 \text{ yr}^{-1}$ derived from Lehodey and Leroy (1999) and using a mean water temperature of 29° C in the study region. Therefore, $M + F = Z \approx P/B$ was 1.670 yr^{-1} .
	Q/B	10-28 qts	Ecopath multi-stanza	Used average daily ration estimate of $4.0\% \text{ BW day}^{-1}$ (Q/B of 14.6 times the population biomass yr^{-1}) which was experimentally determined for small <i>Thunnus albacares</i> (Olson & Boggs, 1986) for “Small yellowfin”, which was the leading stanza in the multi-stanza model in Ecopath. Ecopath estimates Q/B for this group.
	Landings	10-28 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool ($12,086,900 \text{ km}^2$).
	Discards	10-28 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate was divided by the water surface area of the entire Warm Pool ($12,086,900 \text{ km}^2$).
	EE	10-28 qts	Ecopath	Estimated by Ecopath
	Diet	10-28 qts	Allain (2010), Duffy <i>et al.</i> (2017), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for large <i>Thunnus albacares</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>T. albacares</i> in the adjacent tropical Coral Sea (Young <i>et al.</i> 2010).
Juvenile skipjack <i>(Katsuwonis pelamis)</i>	Biomass	1 qt	Hoyle <i>et al.</i> (2011)	Biomass estimate for juvenile skipjack (age class 1 only) in 2005 was taken from Region 2 in the 2011 MFCL stock assessment (Hoyle <i>et al.</i> 2011). The biomass was divided by Region 2's ocean surface area of $25,093,000 \text{ km}^2 = 0.000317 \text{ t km}^{-2}$. The estimate appeared far too low given the small SKJ initial estimate was 0.0235 t km^{-2} . Simon Hoyle explained age 1 is not required in the stock assessment model and any value should be used with caution. Given that the catches are known and the F values are reasonably well known from catch curves of all age classes, Q/B is well known from diet work and bioenergetics models, biomass of this leading stanza was allowed to be estimated by the multi-stanza model within Ecopath, which resulted in a biomass of 0.0160 t km^{-2} .
	P/B	1 qt	Hoyle <i>et al.</i> (2011) Hampton (2000)	Fishing mortality (F) in Region 2 of the 2011 MFCL stock assessment for western Pacific skipjack (Hoyle <i>et al.</i> 2011) for juvenile SKJ (age class 1 quarter) in 2005 was 0.015 yr^{-1} . Natural mortality (M) was estimated from Hampton (2000) where 21-30 cm FL skipjack was $M = 0.8$. The M for Baby SKJ likely to be higher than small SKJ, so $M = 1.0$ was used. Therefore, $M + F = Z \approx P/B$ was 1.015 yr^{-1} . This value was required to be increased by the multi-stanza model to 1.650 yr^{-1} to yield a P/Q close to 0.1 and to reduce EE to around 0.95 while maintaining the reliable F estimate.
	Q/B	1 qt	Ecopath multi-stanza	Used daily ration estimate of $2.18\% \text{ BW day}^{-1}$ (Q/B of 7.957 yr^{-1}) estimated from a quantitative dietary study of the closely related <i>Euthynnus affinis</i> off eastern Australia (Griffiths <i>et al.</i> 2009) for “Small skipjack”, which was the leading stanza in the multi-stanza model in Ecopath. Ecopath estimates Q/B for this group.
	Landings	1 qt	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Catch estimates for juvenile skipjack (age class 1 only) in pole-and-line, PSA and PSU in 2005 was taken from Region 2 in the 2011 MFCL stock assessment (Hoyle <i>et al.</i> 2011). The 2005 estimate was divided by the water surface area of Region 2 of $25,093,000 \text{ km}^2$
	Discards	1 qt	Hoyle <i>et al.</i> (2011)	No discard data available for this size class, assumed to be negligible in all Warm Pool fisheries.
	EE	1 qt	Ecopath	Estimated by Ecopath.
	Diet	1 qts	Allain (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for juvenile <i>Katsuwonis pelamis</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010).

Small skipjack (<i>Katsuwonis pelamis</i>)	Biomass	2-3 qts	Hoyle <i>et al.</i> (2011)	Biomass estimate for small SKJ (age classes 2-3) in 2005 was taken from Region 2 in the 2011 MFCL stock assessment (Hoyle <i>et al.</i> 2011). The biomass was divided by Region 2's ocean surface area of 25,093,000 km ² = 0.0235 t km ⁻² . The multi-stanza model in Ecopath slightly decreased this estimate to 0.0226 t km ⁻² .
	P/B	2-3 qts	Hoyle <i>et al.</i> (2011) Hampton (2000)	Fishing mortality (<i>F</i>) in Region 2 of the 2011 MFCL stock assessment for western Pacific skipjack (Hoyle <i>et al.</i> 2011) for small SKJ (age classes 2-3) in 2005 was 0.257 yr ⁻¹ . Natural mortality (<i>M</i>) was estimated from Hampton (2000) where 21-30 cm FL skipjack was <i>M</i> =0.8. Therefore, <i>M + F = Z</i> ≈ P/B was 1.057 yr ⁻¹ . This value was required to be slightly decreased by the multi-stanza model to 1.00 yr ⁻¹ to yield a P/Q close to 0.1 given the reliable Q/B estimates for skipjack.
	Q/B	2-3 qts	Griffiths <i>et al.</i> (2009)	Used daily ration estimate of 2.18% BW day ⁻¹ (Q/B of 7.957 times the population biomass yr ⁻¹) estimated from a quantitative dietary study of the closely related <i>Euthynnus affinis</i> off eastern Australia (Griffiths <i>et al.</i> 2009).
	Landings	2-3 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Pole-and-line weight estimates were based on observer data in the Warm Pool area and raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that 5% of the SKJ caught in the pole-and-line fishery were Small SKJ. Longline weights were estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	2-3 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Pole-and-line discard biomass estimates were based on observer data in the Warm Pool area and raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that 5% of the SKJ caught in the pole-and-line fishery were Small SKJ. Longline biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	2-3 qts	Ecopath	Estimated by Ecopath
	Diet	2-3 qts	Allain (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for small <i>Katsuwonis pelamis</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Large skipjack (<i>Katsuwonis pelamis</i>)	Biomass	4-16 qts	Hoyle <i>et al.</i> (2011)	Biomass estimate for large SKJ (age classes 4-16) in 2005 was taken from Region 2 in the 2011 MFCL stock assessment (Hoyle <i>et al.</i> 2011). The biomass was divided by Region 2's ocean surface area of 25,093,000 km ² = 0.05638 t km ⁻² . The multi-stanza model in Ecopath slightly decreased this estimate to 0.0401 t km ⁻² .
	P/B	4-16 qts	Hoyle <i>et al.</i> (2011) Hampton (2000)	Fishing mortality (<i>F</i>) in Region 2 of the 2011 MFCL stock assessment for western Pacific skipjack (Hoyle <i>et al.</i> 2011) for large SKJ (age classes 4-16) in 2005 was 0.720 yr ⁻¹ . Natural mortality (<i>M</i>) was estimated from Hampton (2000) where large skipjack was <i>M</i> =1.0. Therefore, <i>M + F = Z</i> ≈ P/B was 0.8698 yr ⁻¹ . This value was required to be slightly increased by the multi-stanza model to 0.878 yr ⁻¹ to yield a P/Q close to 0.1 given the reliable Q/B estimates for skipjack.
	Q/B	4-16 qts	Ecopath multi-stanza	Used daily ration estimate of 2.18% BW day ⁻¹ (Q/B of 7.957 yr ⁻¹) estimated from a quantitative dietary study of the closely related <i>Euthynnus affinis</i> off eastern Australia (Griffiths <i>et al.</i> 2009) for "Small skipjack", which was the leading stanza in the multi-stanza model in Ecopath. Ecopath estimates Q/B for this group.
	Landings	4-16 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Pole-and-line weight estimates were based on observer data in the Warm Pool area and raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that 5% of the SKJ caught in the pole-and-line fishery were Large SKJ. Longline weights were estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).

	Discards	4-16 qts	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Pole-and-line discard biomass estimates were based on observer data in the Warm Pool area and raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. Based on length frequency examination it was assumed that 5% of the SKJ caught in the pole-and-line fishery were Large SKJ. Longline biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	4-16 qts	Ecopath	Estimated by Ecopath
	Diet	4-16 qts	Allain (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for large <i>Katsuwonis pelamis</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Albacore (<i>Thunnus alalunga</i>)	Biomass	All	Hoyle <i>et al.</i> (2012)	Estimate derived from the South Pacific area of the 2012 MFCL stock assessment (Hoyle <i>et al.</i> 2012). The 2005 biomass estimate of 818814.50 t for all age classes combined was divided by the water surface area of the entire warm pool 12,086,900 km ² (since no surface area for the assessment area was provided). The initial value used was 0.06774396 t km ⁻² . But was considered highly uncertain and probably an overestimate since only large albacore are found in the Warm Pool. This was supported during balancing where only 0.0265 t km ⁻² to match the stock assessment's <i>F</i> value and to increase the EE, which was initially far too low (0.03) for a small tuna species with many predators.
	P/B	All	Hoyle <i>et al.</i> (2012)	The 2012 MFCL South Pacific stock assessment for albacore (Hoyle <i>et al.</i> 2012) estimated the average natural mortality (<i>M</i>) as 0.401 yr ⁻¹ and the 2005 fishing mortality (<i>F</i>) as 0.076 yr ⁻¹ . Therefore, $M + F = Z \approx$ P/B was 0.477 yr ⁻¹ . Given the Q/B was reliable, P/B needed to be increased to produce a biologically realistic P/Q of 0.1-0.3. Therefore, P/B was increased to 0.90 yr ⁻¹ to be similar to other tunas in the model.
	Q/B	All	Kirby (2009)	Q/B estimated from a bioenergetics model for south Pacific albacore (Kirby 2009), suggesting Q/B = 14 times the population biomass yr ⁻¹ , averaged across all age classes. Because only adult albacore is common in the Warm Pool, the Q/B value for these older fish should be lower than the average. This value was required to be decreased to 11.00 yr ⁻¹ in order for P/Q to be biologically realistic at around 0.1.
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Allain (2010), Duffy <i>et al.</i> (2017), Young <i>et al.</i> (2010)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for <i>Thunnus alalunga</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>T. alalunga</i> in the adjacent tropical Coral Sea (Young <i>et al.</i> 2010).
Wahoo (<i>Acanthocybium solandri</i>)	Biomass	All	Zischke (2013)	Biomass was estimated using Biomass = catch/fishing mortality. The total catch (4.788E-05 t km ⁻²) and discards (4.837781E-06 t km ⁻²) was divided by an <i>F</i> of 0.74-0.8 estimated by Zischke (2013) for western Pacific wahoo.
	P/B	All	Zischke <i>et al.</i> (2013)	Used von Bertalanffy parameters and the natural mortality empirical equations of Pauly (1980) and Hoenig (1983) using L_{∞} =165.6 cm FL, K =0.360 yr ⁻¹ , t_0 = -3.19 yr ⁻¹ and maximum age 9 years from the study of Zischke <i>et al.</i> (2013) in the Coral Sea with water temperature of 29° C. The two estimates of <i>M</i> were averaged to produce $M = 0.550$ yr ⁻¹ . To calculate P/B of with no available value of <i>F</i> for 2005, it was assumed $F = 0.8M$ if stock is fished at or below $MSY = 0.440$ yr ⁻¹ . Therefore, the initial P/B = 0.990 yr ⁻¹ . However, in order to provide a biologically realistic P/Q value of 0.1-0.3 and given the reasonable Q/B estimate of 17% BW yr ⁻¹ , P/B will need to be much higher. Zischke's work showed the life history of wahoo is similar to dolphinfish, which has a P/B of over 3 (Rivera and Appeldoorn 2000), so P/B was varied until P/Q was around 0.1.
	Q/B	All	Zischke <i>et al.</i> (2013), Palomares and Pauly (1998)	Used von Bertalanffy parameters and the empirical equation of Palomares and Pauly (1998) using L_{∞} =165.6 cm FL and K =0.360 yr ⁻¹ from Zischke <i>et al.</i> (2013) in the Coral Sea and using a water temperature of 29° C to produce an initial Q/B of

				11.7 times the population biomass yr ⁻¹ . However, better information from modelling and field studies exists for other tunas that have far higher Q/B. Wahoo being a high-speed predator would need a high Q/B to maintain metabolic function. Therefore, likely to be at least as high as large YFT for which value is reliable from bioenergetics models. Q/B increased to 17.0 yr ⁻¹ .
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The 2005 estimate for each fishery divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Perelman <i>et al.</i> (2017)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for <i>Acanthocybium solandri</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from a comprehensive quantitative dietary study of <i>A. solandri</i> in the adjacent tropical Coral Sea (Perelman <i>et al.</i> 2017).
Dolphinfish (Mahi Mahi) (<i>Coryphaena hippurus</i>)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate.
	P/B	All	Rivera and Appeldoorn (2000)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) using L_{∞} =145.7 cm FL and K =2.17 yr ⁻¹ from the study of <i>Coryphaena hippurus</i> Rivera and Appeldoorn (2000) in Puerto Rico with water temperature of 29° C to produce M = 1.937 yr ⁻¹ . To calculate P/B of with no available value of F for 2005, it was assumed F = 0.8M if stock is fished at or below MSY = 1.549 yr ⁻¹ . Therefore, the initial P/B = 3.485 yr ⁻¹ .
	Q/B	All	Olson and Galvan-Magana (2002)	Used daily ration estimate of 5.6% BW day ⁻¹ (Q/B of 20.44 times the population biomass yr ⁻¹) estimated from a quantitative dietary study of <i>Coryphaena hippurus</i> in the eastern Pacific (Olson and Galvan-Magana 2002).
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	SPC unpublished dietary data; Pitman and Stinchcomb (2002), Boyle and Limpus (2008).	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for <i>Coryphaena hippurus</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010). Establishment of an additional predation linkage with turtles (as prey (Boyle and Limpus 2008)) and cetaceans (as a predator (Pitman and Stinchcomb 2002)) was necessary based on qualitative or anecdotal evidence.
Small tunas (<i>Euthynnus affinis</i> <i>Auxis thazard</i> <i>Auxis rochei</i> <i>Sarda</i> spp.)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate.
	P/B	All	Kahraman <i>et al.</i> (2011), Valeiras <i>et al.</i> (2008)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) for <i>Auxis rochei rochei</i> (L_{∞} =45.3 cm FL and K =0.397 yr ⁻¹) from Turkish waters (Kahraman <i>et al.</i> 2011), and <i>Euthynnus alletteratus</i> (L_{∞} =90.0 cm FL and K =0.444 yr ⁻¹) from the Seychelles (Valeiras <i>et al.</i> 2008) both with water temperatures of 29° C to produce M = 0.883 yr ⁻¹ and 0.784 yr ⁻¹ , respectively giving an average M = 0.833 yr ⁻¹ . To calculate P/B of with no available value of F for 2005, it was assumed F = 0.8M if stock is fished at or below MSY = 0.667 yr ⁻¹ . Therefore, the initial P/B = 1.500 yr ⁻¹ .

	Q/B	All	Griffiths <i>et al.</i> (2009)	Used daily ration estimate of 2.18% BW day ⁻¹ (Q/B of 7.957 times the population biomass yr ⁻¹) estimated from a quantitative dietary study of <i>Euthynnus affinis</i> off eastern Australia (Griffiths <i>et al.</i> 2009).
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Allain (2010), Griffiths <i>et al.</i> (2009)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for small scombrids caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from a comprehensive quantitative dietary study of <i>Euthynnus affinis</i> in the adjacent tropical Coral Sea (Griffiths <i>et al.</i> 2009).
Escolar & oilfish (<i>Lepidocybium flavobrunneum</i> , <i>Ruvettus pretiosus</i>)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate.
	P/B	All	Unpublished SPC otolith data (Bruno Lehroy - SPC)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) based on a study of 14 escolar otoliths daily aged by Bruno Lehroy (SPC) of fish caught in the Warm Pool. L_{∞} was fixed at 210 cm FL, which was derived from $L_{\max}$ of 200 cm SL in Fishbase; t_0 was fixed at 0, and K estimated as 0.242 yr ⁻¹ . M estimated using a water temperature of 10° C (assumed to be below the mixed layer) $M=0.254$ yr ⁻¹ . To calculate P/B of with no available value of F for 2005, it was assumed $F=0.8M$ if stock is fished at or below $MSY=0.203$ yr ⁻¹ . Therefore, initial P/B = 0.457 yr ⁻¹ .
	Q/B	All	Palomares and Pauly (1998)	Used von Bertalanffy parameters and the empirical equation of Palomares and Pauly (1998) using $W_{\infty}=91676.3$ g with a caudal aspect ratio of 6.55, acting as a carnivore in water temperature of 10° C to produce a Q/B of 3.60 times the population biomass yr ⁻¹ .
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Allain (2010), Keller <i>et al.</i> (2016)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for <i>Lepidocybium flavobrunneum</i> and <i>Ruvettus pretiosus</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from a quantitative dietary study of <i>L. flavobrunneum</i> and <i>R. pretiosus</i> in the western Atlantic Ocean (Keller <i>et al.</i> 2016).
Lancetfish (<i>Alepisaurus brevirostris</i> <i>Alepisaurus ferrox</i>)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate.
	P/B	All	Palomares and Pauly (1998)	Used maximum length recorded on Fishbase.org and used the empirical equation of Palomares and Pauly (1998) to estimate the von Bertalanffy parameter $L_{\infty}=218.5$ yr ⁻¹ and used the average K value from the family level. Natural mortality (M) was then estimated using empirical equation of Pauly (1980) using a water temperature of 10° C (assumed to be below the mixed layer) giving $M=0.120$ yr ⁻¹ . To calculate P/B of with no available value of F for 2005, it was assumed $F=0.8M$ if stock is fished at or below $MSY=0.096$ yr ⁻¹ . Therefore, initial P/B = 0.216 yr ⁻¹ . This value appeared low due to a lack of reliable von Bertalanffy parameter estimates. It was assumed this species should be at least as productive as escolar (0.458 yr ⁻¹), for which more reliable data were available. Therefore, 0.458 yr ⁻¹ was the starting value by predation mortality required P/B to be increased to 0.470 yr ⁻¹ in order to maintain EE at 0.95.
	Q/B	All	Palomares and Pauly (1998)	Used von Bertalanffy parameters and the empirical equation of Palomares and Pauly (1998) using $W_{\infty}=4500.00$ g with a caudal aspect ratio of 1.9, acting as a carnivore in water temperature of 10° C to produce a Q/B of 5.5 times the population biomass yr ⁻¹ . This value was required to be reduced to 4.0 yr ⁻¹ in order for P/Q to be in the vicinity of 0.1.

	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Allain (2010), Young <i>et al.</i> (2010), Choy <i>et al.</i> (2013)	Used diet proportions (in % weight) from comprehensive published and unpublished SPC stomach contents data for <i>Alepisaurus brevirostris</i> and <i>Alepisaurus ferox</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>Alepisaurus</i> spp. in the central Pacific near Hawaii (Choy <i>et al.</i> 2013) and the adjacent tropical Coral Sea (Young <i>et al.</i> 2010).
Opah (<i>Lampris guttatus</i>)	Biomass	All	Griffiths <i>et al.</i> (2009b)	In the absence of biomass estimates for the Warm Pool, used estimate of 0.0049 t km ² of Griffiths <i>et al.</i> (2009b) for opah in the adjacent tropical Coral Sea.
	P/B	All	Francis <i>et al.</i> (2005)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) from the study of <i>Lampris guttatus</i> (Francis <i>et al.</i> 2005) in New Zealand with water temperature of 10° C to produce $M = 0.230 \text{ yr}^{-1}$. To calculate P/B of with no available value of F for 2005, it was assumed $F = 0.8M$ if stock is fished at or below $MSY = 0.184 \text{ yr}^{-1}$. Therefore, the initial P/B = 0.414 yr^{-1} .
	Q/B	All	Palomares and Pauly (1998)	Used maximum length recorded on Fishbase.org and used the empirical equation of Palomares and Pauly (1998) to estimate the von Bertalanffy parameter W_{∞} at a water temperature of 10° C to produce a Q/B of 3.5 times the population biomass yr^{-1} .
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Ecopath	Estimated by Ecopath.
	Diet	All	Allain (2010), Choy <i>et al.</i> (2013)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for <i>Lampris guttatus</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary studies of <i>Lampris</i> spp. in the central Pacific near Hawaii (Choy <i>et al.</i> 2013).
Pomfret (<i>Brama</i> spp. <i>Taractichthys longipinnis</i>)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate.
	P/B	All	Bigelow <i>et al.</i> (1995)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) using $L_{\infty} = 45.6 \text{ cm FL}$ and $K = 1.99 \text{ yr}^{-1}$ to $= 0.02$ from a study in the western and central Pacific Ocean using daily increment counts for <i>Brama japonica</i> (Bigelow <i>et al.</i> 1995). Assumed a water temperature of 10° C to produce $M = 0.8130 \text{ yr}^{-1}$. To calculate P/B with no available value of F for 2005, it was assumed $F = 0.2M$ (0.1626 yr^{-1}) since stock is not a target and is only lightly fished by the distant water longline fleet. Therefore, the initial P/B = 0.9759 yr^{-1} .
	Q/B	All	Palomares and Pauly (1998)	Used von Bertalanffy parameters estimated by Bigelow <i>et al.</i> (1995) and the empirical equation of Palomares and Pauly (1998) using $W_{\infty} = 4269.6 \text{ g}$ with a caudal aspect ratio of 6.55, acting as a carnivore in water temperature of 10° C to produce a Q/B of 8.9 times the population biomass yr^{-1} .
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.

	Diet	All	Allain (2010), Watanabe <i>et al.</i> (2006)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for <i>Brama</i> spp. caught in the Warm Pool and central Pacific Ocean (Allain, 2010) and supplemented with data from comprehensive quantitative dietary study of <i>Brama japonica</i> in the northern central Pacific (Watanabe <i>et al.</i> 2006).
Rainbow runner (<i>Elagatis bipinnulata</i>)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate.
	P/B	All	Iwasaki (1995)	Used von Bertalanffy parameters and the natural mortality empirical equation of Pauly (1980) using $L_{\infty} = 93.0$ cm FL and $K = 0.214$ yr ⁻¹ from Iwasaki (1995) using a water temperature of 29° C to produce $M = 0.4821$ yr ⁻¹ . To calculate P/B with no available value of F for 2005, it was assumed $F = 0.8M$ if stock is fished at or below $MSY = 0.3857$ yr ⁻¹ . Therefore, the initial P/B = 0.8678 yr ⁻¹ .
	Q/B	All	See “Small tunas”	Assumed to be similar to “small tunas”, so the same Q/B value of 7.96 times the population biomass yr ⁻¹ was used.
	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for <i>Elagatis bipinnulata</i> caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Epipelagic crustaceans	Biomass	All	Ecopath	SEAPODYM model estimated a biomass of “epipelagic forage” of 0.110 t km ⁻² in 2005 P/B of 0.84-3.69 yr ⁻¹ for 2005 averaged over 120°E-165°E - 15°N-10°S within the Warm Pool. Based on unpublished survey data, the biomass of this group was broken down into crustaceans (20% - 0.0220 t km ⁻²), fish (65% - 0.0716 t km ⁻²), small fish (5% - 0.0055 t km ⁻²) and molluscs (10% - 0.0110 t km ⁻²) (Valerie Allain – SPC). However, these values were far too low to support the biomass of predators (e.g. tunas) for which reliable empirical data were available. Ecopath left to estimate biomass. Estimates of mysid and decapod biomass off Japan range between 3.0-17.3 t km ⁻² (Kana <i>et al.</i> 2013). The estimate of 10.186 t km ⁻² is therefore reasonable.
	P/B	All	Iguchi and Ikeda (1999)	Estimate of 4.00 yr ⁻¹ used based on P/B estimates of 3.78-5.04 yr ⁻¹ for <i>Euphausia pacifica</i> in the southern Japan Sea (Iguchi and Ikeda 1999).
	Q/B	All	Ikeda and Shiga (1999)	Used published estimates for amphipods (<i>Themisto japonica</i>) in the southern Japan Sea of 6.6% BW d ⁻¹ (Q/B = 24.09 times BW yr ⁻¹) (Ikeda and Shiga 1999), but can range from 6.8-8.2% BW d ⁻¹ for euphausiids (Iguchi and Ikeda 1999).
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Griffiths <i>et al.</i> (2010)	In the absence of quantitative diets of crustaceans from the Warm Pool, used similar prey proportions as a pelagic ecosystem model of eastern Australian waters adjacent to the Warm Pool.
Epipelagic fish (Carangidae spp., Scomberesocidae spp., Exocoetidae spp., Scomber spp.)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate since P/B and Q/B were reliable.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B of 0.84-3.69 yr ⁻¹ for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S. Ecopath required a P/B of 3.4 yr ⁻¹ during balancing, which produce a reasonable P/Q value of 0.243 yr ⁻¹ .
	Q/B	All	Pillar and Barange (1998)	In the absence of Q/B estimates of any taxa in this group for the Warm Pool, used field estimates of daily ration for <i>Trachurus trachurus</i> subtropical waters of South Africa (Pillar & Barange, 1998), 3.8% BW d ⁻¹ (Q/B = 13.87 yr ⁻¹). This value was slightly increased to 14.0 yr ⁻¹ during balancing.

	Landings	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	Discards	All	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line discard estimate were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The 2005 estimate for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²).
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for a range of epipelagic fish species caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Epipelagic small fish (Engraulidae spp., Clupeidae spp., various larval stages of reef fishes)	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate since P/B and Q/B were reliable.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Van der Lingen (1998)	In the absence of Q/B estimates for the Warm Pool, used field estimates of daily ration for <i>Sardinops sagax</i> in the Benguela (Van der Lingen, 1998), which ranged from 2.97-7.58% BW d ⁻¹ for zooplankton-consuming fish. The study region is likely to be more productive than the Warm Pool, so an initial value of 5% BW d ⁻¹ (Q/B = 18.25 yr ⁻¹) was used, which was required to be increased to 22 yr ⁻¹ during balancing.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for a range of epipelagic fish species caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Epipelagic molluscs	Biomass	All	Ecopath	Estimated by Ecopath in the absence of a species-specific biomass estimate since P/B and Q/B were reliable.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Gorbatenko <i>et al.</i> (1995)	Estimated using the average daily ration for two medium-sized pelagic squids in the Bering Sea (4% BW d ⁻¹) resulting in a Q/B of 14.6 yr ⁻¹ .
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Christensen <i>et al.</i> (2009)	Fixed EE value of 0.95 used as recommended by Christensen <i>et al.</i> (2009) for species that have abundant predators.
	Diet	All	Watanabe <i>et al.</i> (2004)	In the absence of dietary studies of epipelagic molluscs in the Warm Pool, used diet composition (by % weight) of the epipelagic neon squid <i>Ommastrephes bartramii</i> in the transitional region of the central North Pacific (Watanabe <i>et al.</i> 2004), which is adjacent to the Warm Pool.
Migratory mesopelagic fish & crustaceans	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Bulman <i>et al.</i> (2006), Palomares and Pauly (1998)	In the absence of Q/B estimates for mesopelagic fishes in the Warm Pool, used von Bertalanffy parameters from published studies and tail aspect ratios from Fishbase.org and used the empirical equation of Palomares and Pauly (1998) to estimate Q/B for <i>Gonostoma elongatum</i> (8.9 yr ⁻¹), <i>Phosichthys argenteus</i> (8.5 yr ⁻¹), <i>Sternoptyx diaphana</i> (14.8 yr ⁻¹) and field study of Myctophidae (6.0 yr ⁻¹) (Bulman <i>et al.</i> 2006). Initially used the average of Q/B values (10.7 yr ⁻¹) which was required to be increased to 11.0 yr ⁻¹ during balancing.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.

	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for a range of epipelagic fish species caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Migratory mesopelagic molluscs	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Gorbatenko <i>et al.</i> (1995)	Estimated using the average daily ration for two medium-sized pelagic squids in the Bering Sea (4% BW d ⁻¹) resulting in a Q/B of 14.6 yr ⁻¹ . Required to be increased slightly to 15.0 yr ⁻¹ during the balancing.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Griffiths <i>et al.</i> (2010)	In the absence of quantitative diets of migratory mesopelagic molluscs from the Warm Pool, used similar prey proportions as “Medium mesopelagic squids” in a pelagic ecosystem model of eastern Australian waters, which is immediately adjacent to the Warm Pool.
Mesopelagic fish & crustaceans	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Bulman <i>et al.</i> (2006), Palomares and Pauly (1998)	In the absence of Q/B estimates for mesopelagic fishes in the Warm Pool, used von Bertalanffy parameters from published studies and tail aspect ratios from Fishbase.org and used the empirical equation of Palomares and Pauly (1998) to estimate Q/B for <i>Gonostoma elongatum</i> (8.9 yr ⁻¹), <i>Phosichthys argenteus</i> (8.5 yr ⁻¹), <i>Sternoptyx diaphana</i> (14.8 yr ⁻¹) and field study of Myctophidae (6.0 yr ⁻¹) (Bulman <i>et al.</i> 2006). Initially used the average of Q/B values (10.7 yr ⁻¹) which was required to be increased to 13.0 yr ⁻¹ during balancing.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for a range of epipelagic fish species caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Mesopelagic molluscs	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Gorbatenko <i>et al.</i> (1995)	Estimated using the average daily ration for two medium-sized pelagic squids in the Bering Sea (4% BW d ⁻¹) resulting in a Q/B of 14.6 yr ⁻¹ . Required to be decreased slightly to 13.0 yr ⁻¹ during the balancing.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Griffiths <i>et al.</i> (2010)	In the absence of quantitative diets of mesopelagic molluscs from the Warm Pool, used similar prey proportions as “Small mesopelagic squids” (non-migratory) in a pelagic ecosystem model of eastern Australian waters, which is immediately adjacent to the Warm Pool.
Highly migratory bathypelagic forage	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Griffiths <i>et al.</i> (2010)	In the absence of Q/B estimates, used an initial value of 8.70 yr ⁻¹ used for “Medium mesopelagic fishes” in a pelagic ecosystem model of eastern Australian waters adjacent to the Warm Pool. Reduced to 8.00 yr ⁻¹ during balancing to reduce predation pressure on zooplankton groups.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath

	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for a range of epipelagic fish species caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Migratory bathypelagic forage	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Griffiths <i>et al.</i> (2010)	In the absence of Q/B estimates, used an initial value of 8.70 yr ⁻¹ used for “Medium mesopelagic fishes” in a pelagic ecosystem model of eastern Australian waters adjacent to the Warm Pool. Reduced to 8.00 yr ⁻¹ during balancing to reduce predation pressure on zooplankton groups.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for a range of epipelagic fish species caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Bathypelagic forage	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Griffiths <i>et al.</i> (2010)	In the absence of Q/B estimates, used an initial value of 8.70 yr ⁻¹ from a pelagic ecosystem model of eastern Australian waters adjacent to the Warm Pool. Reduced to 6.60 yr ⁻¹ as the group occupies cold waters ration is assumed to be significantly lower due to lower metabolic demands, but also to reduce predation pressure on zooplankton groups.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Allain (2010)	Used diet proportions (in % weight) from unpublished SPC stomach contents data for a range of epipelagic fish species caught in the Warm Pool and central Pacific Ocean (Allain, 2010).
Mesozooplankton	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Froneman <i>et al.</i> (1996)	Very few data available in the literature, especially for the western Pacific. Therefore, in the absence of local estimates, used the estimated daily ration of <i>Salpa thompsoni</i> (72.7% BW d ⁻¹) resulting in a Q/B of 265.36 yr ⁻¹ . Required to be decreased 218.0 yr ⁻¹ during model balancing.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	Kjørboe (1997), Schnetzer and Steinberg (2002)	Used diet proportions inferred from studies of oceanic mesozooplankton described by Kjørboe (1997) and Schnetzer & Steinberg (2002) in the tropical Atlantic Ocean. Given the coarse resolution of prey available to mesozooplankton in the Warm Pool model, these generalised diets are likely to be representative in the Warm Pool at the functional group level.
Microzooplankton	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	Burkill <i>et al.</i> (1993)	Very few Q/B estimates available in the literature, especially for the western Pacific. Therefore, in the absence of local estimates, used the estimated daily ration for microzooplankton in the northeastern Atlantic, which consumed 100-800% of their BW d ⁻¹ . Their study area is nutrient rich, unlike the Warm Pool and therefore, it was assumed that daily ration is probably at the lower end of their estimates. Thus, an initial value of 100% BW d ⁻¹ (Q/B = 365.00 yr ⁻¹) was used, which required no change during balancing.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath

	Diet	All	Calbet (2008)	Used diet proportions inferred from studies of oceanic microzooplankton described Calbet (2008). Given the coarse resolution of prey available to mesozooplankton in the Warm Pool model, these diet generalisations are likely to be representative in the Warm Pool at the functional group level.
Large phytoplankton	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	N/A	Not applicable to primary producers.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	N/A	Not applicable to primary producers.
Small phytoplankton	Biomass	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated biomass for 2005 averaged over the Warm Pool area 110°E-180°W - 15°N-15°S.
	P/B	All	Lehodey <i>et al.</i> (2010)	SEAPODYM model estimated P/B for 2005 in the Warm Pool area 110°E-180°W - 15°N-15°S.
	Q/B	All	N/A	Not applicable to primary producers.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	N/A	Not applicable to primary producers.
Detritus	Biomass	All	Emerson <i>et al.</i> (1987)	There are no recent biomass estimates for detritus in the open ocean of the Warm Pool, apart from those of Emerson <i>et al.</i> (1987) for the abyssal plains of three open ocean regions, which includes the equatorial Pacific. These ranged from 53-103 t km ⁻² . Given the Warm Pool is oligotrophic, we used the minimum estimate (53 t km ⁻²) as an initial value, which did not require manipulation during balancing.
	P/B	All	N/A	Not applicable to non-living detrital groups.
	Q/B	All	N/A	Not applicable to non-living detrital groups.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	N/A	Not applicable to non-living detrital groups.
Fishery discards	Biomass	All	Ecopath “Discards” input table	Taken from the total from the “Discards” table in Ecopath summing all discards from all fisheries divided by the Warm Pool model area of 12,555,000 km ² which was initially 0.002720543 t km ⁻²
	P/B	All	N/A	Not applicable to non-living detrital groups.
	Q/B	All	N/A	Not applicable to non-living detrital groups.
	Landings	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	Discards	All	N/A	Not captured in any of the four fisheries included in the Warm Pool model.
	EE	All	Ecopath	Estimated by Ecopath
	Diet	All	N/A	Not applicable to non-living detrital groups.

Table S2. Initial diet matrix used in the Warm Pool Ecopath model prior to balancing. Note the absence of seabirds and cetaceans, which were added after the initial model was balanced.

Group No.	Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	Turtles							0.00001		0.00001	0.00001	0.00001								
2	Small swordfish							0.00092		0.00077	0.00099	0.00106								
3	Large swordfish							0.00046		0.00039	0.00049	0.00053								
4	Blue marlin				0.00867			0.00046		0.00039	0.00049	0.00053								
5	Striped marlin							0.00046		0.00039	0.00049	0.00053								
6	Other billfish					0.02706		0.00046		0.00039	0.00049	0.00053				0.00059				
7	Mako shark																			
8	Blue shark																			
9	Silky shark																			
10	White tip shark																			
11	Other sharks										0.00493									
12	Small bigeye tuna				0.00198	0.07603	0.05513	0.04618		0.00386	0.00985	0.01058		0.01418		0.00971		0.01285	0.00830	
13	Large bigeye tuna									0.00493	0.00482									
14	Small yellowfin tuna				0.00181	0.07603	0.03952	0.18473		0.00386	0.00985	0.01058		0.00899		0.02751			0.01349	
15	Large yellowfin tuna									0.00493	0.00529									
16	Juvenile skipjack					0.10271	0.08820						0.00056	0.07262	0.10542	0.12468		0.07767	0.14835	
17	Small skipjack				0.25514	0.07603	0.21553	0.09237	0.10610	0.07712	0.09851	0.01058		0.19526		0.23931				
18	Large skipjack				0.54469			0.36087		0.23137	0.19701	0.00846								
19	Albacore				0.00297	0.07603	0.01629	0.05314		0.00386	0.00493	0.00481				0.05463				
20	Wahoo					0.03801		0.00189		0.00386	0.00493	0.00529								
21	Dolphinfish				0.02114	0.03801	0.09188		0.00531	0.00386	0.00493	0.00529								
22	Small tunas				0.00889	0.07603	0.08264	0.00189		0.03856	0.09851	0.01058		0.01961	0.00575	0.01961		0.02087	0.05153	0.01961
23	Escolar / Oilfish		0.00908	0.00989						0.00386	0.00493	0.00479		0.01961	0.00043	0.00311		0.00094		
24	Lancetfish		0.09173	0.09172	0.09091	0.09091	0.09091	0.17982	0.31830	0.08658	0.29552	0.09427								
25	Opah							0.00189		0.00173		0.00188								
26	Pomfret		0.00908	0.00989		0.00281			0.00531		0.00985	0.01058	0.01366	0.04128	0.01187	0.11359		0.00141	0.01001	0.01349
27	Rainbow runner				0.00889				0.00531	0.00386	0.00985	0.01058			0.03663					
28	Epi. crustaceans	0.08850		0.00015	0.00002	0.00033	0.00088	0.00007	0.00011	0.00003			0.01418	0.00420	0.11084	0.02401	0.08333	0.06906	0.02354	0.11273
29	Epi. fish	0.08850			0.00157		0.06575	0.01427	0.15915	0.13293	0.09851	0.42302			0.05188	0.04456	0.04167	0.05692	0.03446	0.03924
30	Epi. small fish	0.17699	0.08272		0.03651	0.05051	0.14284	0.01154	0.03450	0.27034			0.03754	0.00685	0.48481	0.11881	0.53571	0.66479	0.29298	0.17279
31	Epi. molluscs	0.29204	0.00172	0.29663	0.00074	0.00114		0.00087	0.15915	0.07712	0.08750	0.21151	0.05890	0.04624	0.01653	0.05125	0.29762	0.00488	0.00318	0.10990
32	Migr. meso. fish / crust.		0.00288	0.00414			0.02703		0.10610				0.00956	0.00545	0.00756	0.01850		0.01411	0.03184	0.02721
33	Migr. meso. moll		0.10055	0.14540	0.01589		0.00554		0.05305			0.01058	0.08707	0.10610	0.08152	0.03686		0.05856	0.28451	0.01590
34	Meso. fish / crust.		0.34758	0.15589		0.00554	0.04412	0.00005				0.05288	0.36910	0.34640	0.03200	0.05845		0.01013	0.08716	0.26193
35	Meso. molluscs			0.00545		0.26281	0.02684					0.05288	0.01611	0.00872	0.00111	0.02937		0.00557	0.00785	0.09746
36	H. migr. bathy. forage		0.21089	0.25528									0.08153	0.04119	0.01315	0.00446		0.00220	0.00102	0.05799
37	Migr. bathy. forage												0.00302	0.00925		0.00016				0.00434
38	Bathypelagic forage		0.14376	0.02557	0.00016		0.00689			0.00728			0.30877	0.07249	0.04051	0.02083		0.00086	0.00075	0.06736
39	Mesozooplankton	0.35400															0.04167	0.00012	0.00008	0.00004
40	Microzooplankton																			
41	Large phytoplankton																			
42	Small phytoplankton																			
43	Detritus																			
44	Fishery discards							0.04762	0.04762	0.04762	0.04762	0.04762								

Table S2 continued.

Group No.	Prey \ predator	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
1	Turtles		0.00010																			
2	Small swordfish	0.00020	0.00427																			
3	Large swordfish																					
4	Blue marlin																					
5	Striped marlin																					
6	Other billfish																					
7	Mako shark																					
8	Blue shark																					
9	Silky shark																					
10	White tip shark																					
11	Other sharks																					
12	Small bigeye tuna																					
13	Large bigeye tuna																					
14	Small yellowfin tuna																					
15	Large yellowfin tuna																					
16	Juvenile skipjack	0.12312	0.25221						0.06354													
17	Small skipjack	0.23694																				
18	Large skipjack																					
19	Albacore																					
20	Wahoo																					
21	Dolphinfish		0.09813																			
22	Small tunas	0.01961		0.01961																		
23	Escolar / Oilfish	0.02604																				
24	Lancetfish																					
25	Opah																					
26	Pomfret	0.01614			0.07286	0.01198																
27	Rainbow runner																					
28	Epi. crustaceans	0.00120	0.00258	0.02610	0.01540	0.01854	0.02021	0.00627	0.03419		0.08055	0.08929	0.14793	0.16955	0.19900			0.27506				
29	Epi. fish	0.02715	0.21201			0.01486			0.44240					0.09080	0.09950			0.06876				
30	Epi. small fish	0.37394	0.31760	0.90727	0.00729	0.10775			0.41620		0.70558		0.14793	0.17943	0.19900	0.00435		0.06876				
31	Epi. molluscs	0.00182	0.00498	0.00157	0.02201	0.19769		0.22090	0.00723		0.08159	0.01786		0.11330	0.29851			0.10108				
32	Migr. meso. fish / crust.	0.02871	0.00245	0.00010		0.01844		0.01098	0.00013				0.19752		0.14925	0.41513	0.19510	0.03438	0.11056	0.06434		
33	Migr. meso. moll	0.04635	0.03734	0.01350	0.04059	0.00336	0.15810		0.00101				0.09918			0.16605	0.19510	0.03438	0.11056			
34	Meso. fish / crust.	0.03418	0.04928	0.02127		0.49210	0.45560	0.52854	0.00019							0.17968	0.29265		0.33167	0.11490		
35	Meso. molluscs	0.04399			0.65001	0.05175										0.16605	0.01951		0.16584			
36	H. migr. bathy. forage		0.01841	0.01058		0.00523	0.23955	0.06831	0.03503				0.19752			0.02098				0.43662		
37	Migr. bathy. forage				0.18781	0.03149		0.01882									0.09755			0.13328		
38	Bathypelagic forage	0.02061	0.00060		0.00405	0.04677	0.12625	0.14618												0.04596		
39	Mesozooplankton		0.00015			0.00002	0.00029		0.00006	0.45000	0.12238	0.80357	0.16793	0.38570	0.04975	0.04278	0.19510	0.41259	0.27640			
40	Microzooplankton									0.10000			0.04198	0.05625							0.20000	
41	Large phytoplankton									0.05000	0.00990	0.08929									0.10000	0.10000
42	Small phytoplankton																				0.05000	0.80000
43	Detritus									0.40000											0.19992	0.65000
44	Fishery discards													0.00498	0.00498	0.00498	0.00498	0.00498	0.00498	0.00498		0.10000

Table S3. Final diet matrix used in the Warm Pool Ecopath model after balancing.

Group No.	Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	Seabirds									0.00001		0.00001	0.00001	0.00001						
2	Cetaceans									0.00050		0.00050	0.00050	0.00500						
3	Turtles	0.00010								0.00010		0.00380	0.00400	0.00800						
4	Small swordfish									0.00092		0.00077	0.00099	0.00106						
5	Large swordfish									0.00046		0.00038	0.00049	0.00053						
6	Blue marlin									0.00055		0.00060	0.00070	0.00080						
7	Striped marlin									0.00046		0.00038	0.00049	0.00053						
8	Other billfish									0.00046		0.00038	0.00049	0.00053						
9	Mako shark											0.00001	0.00001	0.00001						
10	Blue shark									0.00040		0.00001	0.00001	0.00001						
11	Silky shark									0.00010			0.00001	0.00001						
12	White tip shark									0.00010				0.00001						
13	Other sharks									0.00001		0.00001	0.00493							
14	Small bigeye tuna				0.00300	0.02100	0.03000	0.03500	0.03100	0.05900	0.01000	0.02000	0.03800	0.02500		0.00400		0.01000		
15	Large bigeye tuna					0.02000	0.00800			0.07100		0.00500	0.02100	0.02100						
16	Small yellowfin tuna					0.06100	0.13500	0.06400	0.09000	0.16900	0.01600	0.02000	0.07500	0.14500		0.05000		0.05500		
17	Large yellowfin tuna						0.00900			0.10000		0.00500	0.00650	0.00620						
18	Juvenile skipjack	0.00100	0.00100					0.08000	0.07000						0.00056	0.07390	0.01300	0.07230		0.00001
19	Small skipjack		0.00030				0.19300	0.08026	0.20100	0.06100	0.01500	0.02000	0.10100	0.01056		0.08000	0.00400	0.08500		
20	Large skipjack						0.25500	0.00528	0.00521	0.12600		0.02000	0.17100	0.00845		0.05000		0.03000		
21	Albacore				0.00300	0.05000	0.04500	0.08026	0.06000	0.17500	0.03000	0.01000	0.04200	0.06300		0.01100		0.00100		
22	Wahoo						0.00998	0.01000		0.00189		0.00100	0.01000	0.00800						
23	Dolphinfish	0.00100	0.00600				0.04988	0.04013	0.09578	0.00500	0.00100	0.02000	0.00493	0.00528		0.00100	0.01000	0.01033		
24	Small tunas	0.00100	0.00800				0.10900	0.08400	0.05100	0.05000		0.04000	0.05400	0.00100		0.02200	0.01000	0.08000		0.03142
25	Escolar / Oilfish				0.00908	0.00989	0.00100					0.00100	0.00493	0.00478		0.00118	0.00043	0.00321		
26	Lancetfish				0.09173	0.09172	0.09064	0.09597	0.09477	0.12900	0.02000	0.00300	0.23500	0.09415				0.01033		
27	Opah						0.00100			0.00189		0.00172	0.00100	0.00188						
28	Pomfret	0.00100			0.00908	0.00989	0.00100	0.00297			0.00531	0.00100	0.00985	0.01056	0.01366	0.05500	0.01188	0.14900		0.00143
29	Rainbow runner						0.00887		0.00130		0.00531	0.00100	0.00985	0.01056			0.03665	0.03200		
30	Epi. crustaceans	0.04000	0.04000	0.08850							0.00011	0.05000			0.01418	0.00010	0.16100	0.00102	0.08597	0.11147
31	Epi. fish	0.15000	0.20100	0.08850			0.00157	0.06700	0.07600	0.01100	0.06000	0.10600	0.09600	0.20400			0.05400	0.04603	0.04298	0.06200
32	Epi. small fish	0.15000	0.26500	0.17699	0.08272		0.03641	0.07180	0.11100	0.01154	0.03451	0.20100		0.01400	0.03754	0.00697	0.49506	0.14200	0.55266	0.69817
33	Epi. molluscs	0.54600	0.39400	0.29204	0.00172	0.27600	0.00073	0.01000	0.01100	0.02003	0.54000	0.36200	0.09702	0.23400	0.05889	0.06000	0.01654	0.07234	0.30703	0.00495
34	Migr. meso. fish / crust.	0.05000	0.01912		0.00288	0.00414			0.02818		0.10611	0.02000			0.36000	0.00555	0.04800	0.01912		0.02331
35	Migr. meso. moll	0.05000	0.02000		0.10055	0.10400	0.01450				0.05305	0.03500			0.18000	0.12000	0.09005	0.03808		0.04924
36	Meso. fish / crust.		0.02000		0.34100	0.10600		0.00585	0.04600	0.00005		0.02000		0.05281	0.25900	0.30500	0.03502	0.06600		0.01027
37	Meso. molluscs		0.01200			0.00545		0.26700	0.02798		0.09900	0.02000		0.05281	0.01611	0.01000	0.00111	0.04134		0.00564
38	H. migr. bathy. forage		0.00600		0.21089	0.22000									0.05000	0.06100	0.01316	0.00460		0.00223
39	Migr. bathy. forage				0.10000	0.01600									0.01000	0.00942		0.00017		
40	Bathypelagic forage				0.04400	0.00500										0.07377		0.03100		
41	Mesozooplankton			0.35400															0.01135	0.00012
42	Microzooplankton																			
43	Large phytoplankton																			
44	Small phytoplankton																			
45	Detritus																			
46	Fishery discards	0.01000	0.00800							0.00500	0.00500	0.01000	0.01000	0.01000						

Table S3 continued.

Group No.	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
1																							
2																							
3				0.00010																			
4																							
5																							
6																							
7																							
8																							
9																							
10																							
11																							
12																							
13																							
14			0.01000	0.00100																			
15																							
16			0.12100	0.03000																			
17																							
18	0.00100		0.04000	0.04000						0.01000													
19			0.14500	0.03000																			
20			0.00010																				
21																							
22						0.07000																	
23			0.08000																				
24	0.07384	0.01961	0.05000	0.02400																			
25			0.00500							0.09000													
26						0.02000			0.10400														
27						0.00001																	
28	0.01057	0.01349	0.01617			0.07286	0.01198	0.20100															
29			0.02100			0.00500	0.00100																
30	0.04221	0.11273		0.10200	0.03000	0.00100	0.03000	0.02021	0.00627	0.03437										0.20300	0.01900		
31	0.08100	0.03924	0.05900	0.27500	0.00600	0.04000	0.01486		0.00800	0.47000										0.06000			
32	0.37782	0.17279	0.29700	0.35200	0.92298	0.04000	0.10775		0.02500	0.44200										0.06000			
33	0.00336	0.10990	0.00183	0.03300	0.00157	0.18200	0.19769	0.06000	0.08000	0.00727										0.08000			
34	0.05277	0.03700	0.02877	0.05087	0.00010	0.04250	0.02700	0.11200	0.01098	0.00013										0.03200	0.12000	0.03000	
35	0.25599	0.01590	0.04645	0.03854	0.01350	0.02000	0.00800	0.08900	0.14000	0.00102										0.09800	0.02600		
36	0.09198	0.26193	0.03425	0.00400	0.02127	0.05700	0.48000	0.15300	0.41000	0.00019										0.18000	0.08000		
37	0.00829	0.09746	0.04408			0.25800	0.05175	0.11300	0.22200											0.10300	0.03000		
38	0.00107	0.07000		0.01900	0.00500	0.07000	0.01000	0.04000	0.02300	0.03521										0.05000	0.25300		
39		0.01000				0.11800	0.04000	0.00120	0.00200											0.01000	0.17100		
40		0.04000				0.00405	0.02000													0.05000			
41	0.00009	0.00004		0.00016			0.00002	0.01620	0.07300	0.00006	0.14500	0.12000	0.30500	0.12000	0.34000	0.04975	0.19000	0.19510	0.40100	0.28000	0.12000		
42											0.19500		0.13000	0.06000	0.04000		0.03600					0.16000	
43											0.09000	0.00990	0.14000	0.01000			0.01000		0.01000	0.01000		0.08000	0.06000
44											0.08000		0.09000								0.05000	0.70000	
45											0.49000	0.19000	0.18000		0.11000		0.14900		0.12000	0.13000	0.29000	0.71000	0.24000
46																0.00100		0.00100		0.00001	0.00001		

Table S4. Fishery landings and discards (in t km⁻²) in 2005 for each functional group within each of the four fisheries defined in the Warm Pool Ecopath model.

Group No.	Functional group name	Landings (1.0 x 10 ⁶ t km ⁻²)					Discards (1.0 x 10 ⁶ t km ⁻²)				
		LL	PSA	PSU	PL	Total	LL	PSA	PSU	PL	Total
1	Seabirds						0.461				0.461
2	Cetaceans						9.100	0.131	0.112		9.343
3	Turtles						0.300	0.100	0.100		0.500
4	Small swordfish	12.800				12.800					
5	Large swordfish	172.514				172.514					
6	Blue marlin	1.900	13.100	9.570		33.570					
7	Striped marlin	13.000	2.850	1.480		17.330					
8	Other billfish	25.800	14.600	6.290		46.690	5.300	4.927	2.867		13.942
9	Mako shark	9.000	0.622	0.140		9.146	8.480				8.480
10	Blue shark	87.000				87.000					
11	Silky shark	19.800	58.600	16.700		95.100	49.400	5.652	1.962	0.752	111.719
12	White tip shark	21.600	2.560	0.325		24.485	1.890	8.542	0.263		1.692
13	Other sharks	33.600	0.992	0.120		34.712	32.800				32.800
14	Small bigeye tuna	321.222	221.535	88.700	0.377	2431.834				0.622	0.622
15	Large bigeye tuna	432.814	1.200	63.700	0.877	57.591					
16	Small yellowfin tuna	112.170	9545.440	32.811	12.100	13662.370				1.534	1.534
17	Large yellowfin tuna	717.430	89.967	159.500	1.910	3199.970					
18	Juvenile skipjack	0.100	0.772	0.682	6.900	8.354					
19	Small skipjack	0.630	337.348	241.956	2125.856	5738.166				1.396	1.396
20	Large skipjack	0.356	14653.850	1579.790	373.434	28964.430				1.188	1.188
21	Albacore	218.751				218.751					
22	Wahoo	45.000	2.520	0.360		47.880	3.400	1.196	0.118	0.300	4.838
23	Dolphinfish	1.900	73.300	2.910	0.620	87.172	0.542	25.830	1.518	0.300	27.723
24	Small tunas	27.600	65.200	1.300	0.316	13.416		81.358	3.637	0.300	84.973
25	Escolar & oilfish	1.970				1.970	3.400				3.400
26	Lancetfish	1.990				1.990	4.550				4.550
27	Opah	24.100				24.100	3.200				3.200
28	Pomfret	2.490				2.490	1.770	0.277	0.312		2.780
29	Rainbow runner	1.310	256.730	7.110	4.200	269.323	0.567	724.838	2.111	0.600	746.116
30	Epipelagic crustaceans										
31	Epipelagic fish	2.610	27.923	1.000	0.120	22.653	3.170	27.923	9.751	0.300	221.145
32	Epipelagic small fish										
33	Epipelagic molluscs										
34	Migratory mesopelagic fish & crustaceans										
35	Migratory mesopelagic molluscs										
36	Mesopelagic fish & crustaceans										
37	Mesopelagic molluscs										
38	Highly migratory bathypelagic forage										
39	Migratory bathypelagic forage										
40	Bathypelagic forage										
41	Mesozooplankton										
42	Microzooplankton										
43	Large phytoplankton										
44	Small phytoplankton										
45	Detritus										
46	Fishery discards										

Appendix S2 – Balancing the Ecopath model

Balancing the Ecopath model

Once all model parameters were estimated from the available datasets and literature for 2005, and multi-stanza models were constructed for swordfish, bigeye tuna, yellowfin tuna and skipjack, balancing began by first focusing on the 14 functional groups for which high quality age structured stock assessments had been undertaken.

The model was attempted to be balanced to inspect the preliminary ecotrophic efficiency (EE) and production/consumption estimates (P/Q) estimated by Ecopath for these key species. For most living groups, the primary aim during balancing is to allow Ecopath to estimate an EE of around 0.95 for species that are either heavily fished, or are expected to experience high predation rates (Christensen *et al.* 2009). A value of around 0.95 indicates that 95% of the biomass of the group is utilised within the system. In contrast, highly abundant and/or highly productive species (e.g. zooplankton), or high trophic level predators that have very few natural predators are expected to have low EE values of 0.1–0.3. This indicates that only a small percentage of the species' biomass is utilised within the system, while the remainder is transferred to detritus or exported from the system.

In cases where the EE value was unrealistic—for example juvenile skipjack initially had an EE of over 1000—the biomass estimate was checked, followed by an inspection of the predation mortality and fishing mortality estimates calculated by Ecopath. Commonly, the most scientifically defensible option to reduce the EE of a group—when all other parameters appear plausible—is to slightly reduce the proportion it contributes to the diets of predators that have a very high biomass or are highly productive. The converse applies where the EE is too low, requiring greater predation mortality when all other parameter values are considered valid. These are common approaches used by Ecopath practitioners since the diet composition

of most marine predators is often highly variable in space and time and an Ecopath model attempts to capture this variability for a single year. In many respects the initial diet composition of a group should be viewed as a reasonable starting point with an expectation of fine tuning using knowledge of the species and the system (see Christensen *et al.* 2009).

After the EE values were considered reasonable, the P/Q estimate for each group was checked to ensure it was biologically realistic. In most cases, P/Q should be between 0.1 and 0.3. In cases where P/Q was unrealistic, such as for Wahoo initially having a P/Q of 0.001, the P/B and Q/B values were inspected and revised appropriately. It was often the case that daily ration estimates from the literature were underestimated or unreliable, while the P/B value was underestimated due to poor von Bertalanffy growth parameters from which natural mortality (M) was estimated, or a lack of fishing mortality information, which was generally the case for bycatch species. Further searches of the literature or sourcing unpublished work or expert opinion often provided more reliable information that assisted in producing more reasonable P/B estimates.

Once the aforementioned process was completed for the groups having the most reliable information, and then applied sequentially to prey of these groups occupying lower trophic levels, and so on.

Diagnostics of Ecopath model validity and stability

On completion of the balancing process, we followed ‘best practice’ (Heymans *et al.* 2016) and “PREBAL” (Link 2010) diagnostics to test the biological and ecological validity and overall stability of the balanced model, which is summarised in Figure S1. The primary ‘rule of thumb’ proposed by Link (2010) is that biomass of functional groups in the model should span 5-7 orders of magnitude and decline at a rate of around 5-10% with increasing trophic level. Figure S1a shows the biomass of functional groups in the Warm Pool model ranging

from $0.000057 \text{ t km}^{-2}$ to 10.95 t km^{-2} —slightly more than 6 orders of magnitude. Biomass decreased with increasing trophic level at a rate of 22%, with values of the majority of groups being close to the regression line.

Both P/B and Q/B values are expected to decline with increasing trophic level (Link 2010, Heymans *et al.* 2016), which was the case for both parameters in the Warm Pool model (Figure 1Sb and 1Sc).

Finally, the gross food conversion efficiency (P/Q) of each functional group should only produce a fraction of the biomass it has consumed (Heymans *et al.* 2016), usually on the order of 0.05-0.3 (Christensen *et al.* 2009). All functional groups had P/Q values of between 0.05-0.3 with the exception of four groups, which represent special cases. *Seabirds* and *Sea turtles* had a P/Q of 0.01 and 0.026, respectively since their production is quite low. Seabirds, such as albatross, have a very low natural mortality rates (i.e. P/B) (Tuck *et al.* 2001) due to high parental investment, but have high consumption rates (Q/B) to satisfy their high metabolic demands, thus resulting in a very low P/Q value. Similarly, *Sea turtles* are long-lived and also have a low natural mortality, but have a reasonably high Q/B due to the consumption of low energy prey, primarily gelatinous zooplankton. In contrast, “*Mesopelagic fish and crustaceans*” and “*Migratory mesopelagic fish and crustaceans*” had slightly higher than expected P/Q values (0.32-0.34) owing to their very high productivity and biomass that is required to sustain very high predation mortality from a range of open ocean predators as a ‘wasp waist’ group (Griffiths *et al.* 2013), as is indicated by the high EE values for these groups of 0.988 and 0.993, respectively.

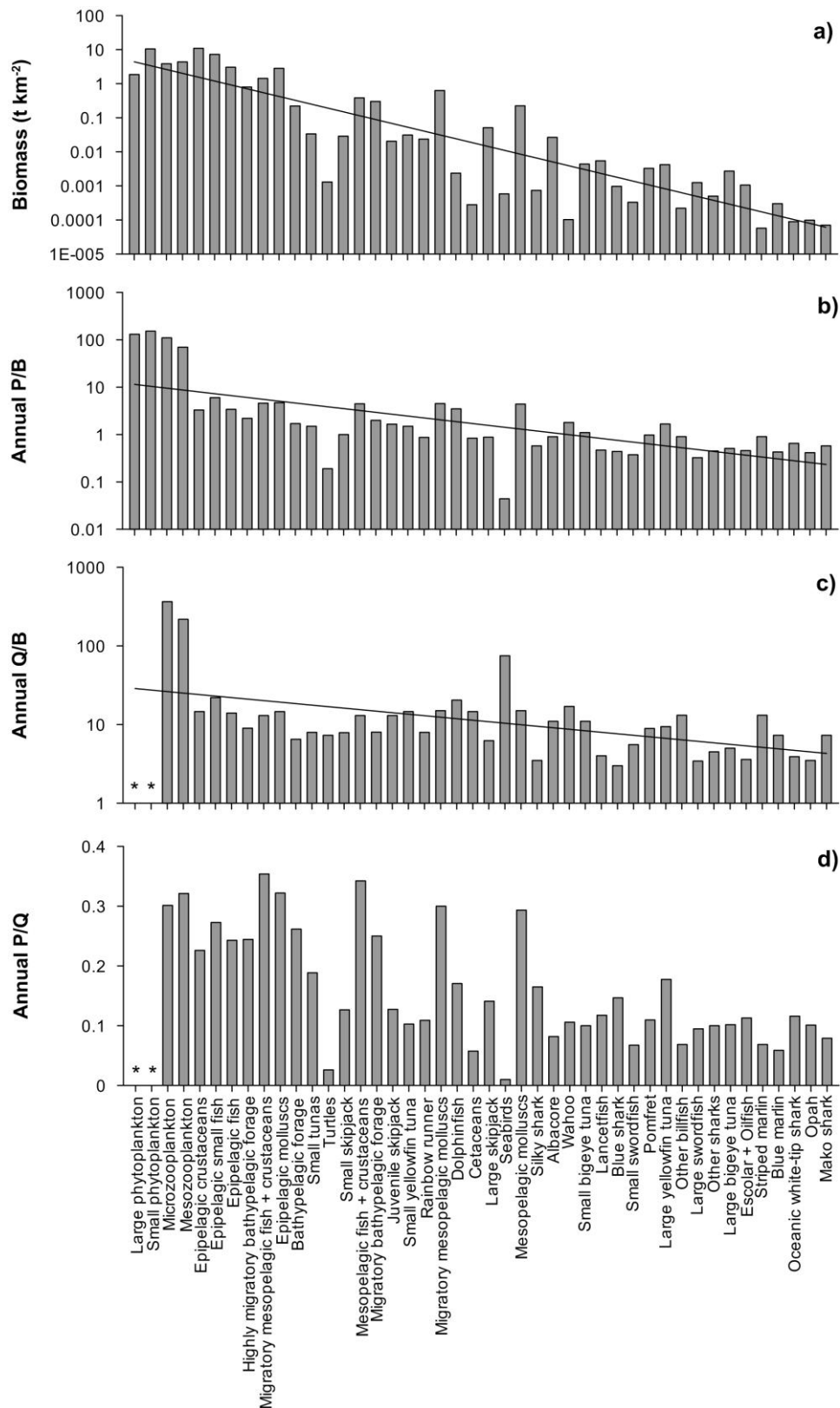


Figure S1. PREBAL diagnostics showing a) biomass, b) P/B, c) Q/B, and P/Q values for each living functional group in the balanced Warm Pool Ecopath model. Asterisks denote primary producer groups for which a Q/B is not applicable.

Once we confirmed that the underlying biological and ecological assumptions of the Ecopath model had been satisfied using the recommendations of Link (2010) and Heymans *et al.* (2016), the thermodynamic stability of the Warm Pool model was assessed. The model was transferred to Ecosim and run for 1000 years in the absence of fishing to assess the model's temporal stability and to observe whether any group went extinct, increase indefinitely, or demonstrate erratic or oscillatory trends in their biomass. The model produced a stable 'flat line' for all functional groups with no indication of temporal instability. These diagnostic tests confirmed that the Ecopath model was robust and was ready to be used for forecasting changes to the structure of the Warm Pool ecosystem under specific fishery and/or environmental scenarios specified in Ecosim.

Appendix S3 – Calibrating the Ecosim model

Fitting the Ecosim model to time series data

Once the Ecopath model was confirmed to be internally stable and ecologically realistic, we sought to optimise the reliability of predictions in Ecosim scenario modelling by calibrating the model to 111 data time series consisting of biomass, and/or fishing mortality and/or catch data for 37 of the 44 living functional groups. The data types used and their sources are detailed in Table S5. Although several data time series for economically important species extended from 1952 to 2010, data for years prior to around 1980 were considered unreliable due to various data reporting issues. Therefore, we used the years 1952–1980 for essentially 'burning in' the Ecosim model but focus on the years after 1980.

The data for the 37 functional groups were imported into Ecosim and weighted according to the perceived reliability of the data and its relative importance in driving the model fitting. The 30 most sensitive interactions between predators and prey were identified using the non-

linear search procedure within Ecosim. The prey vulnerability rate (v , see Equation 4) for the most sensitive predator-prey interactions were then iteratively adjusted by 1% until the sums of squares error (SS) was minimised to produce the ‘best’ model fit to the time series data. This process was repeated 20 times—with v values being reset before each iteration—to minimise the chance of the nonlinear search procedure becoming ‘trapped’ in local parameter minima. The matrix of vulnerability values by predator/prey interaction that contributed to the fit with the lowest SS from the 20 iterations was saved as the ‘optimal’ vulnerability matrix, which is shown in Table S6.

In searching for the best combination of vulnerability values, the time series data were linked to an estimated trend of primary productivity anomalies, in the form of a forcing function, forced upon the “Large phytoplankton” group. A variance value of 10 was used for the model fitting, implying that the model should attempt to capture any abrupt variability in the biomass and catch data through time. This procedure did not reduce the SS and so the forcing function was removed and not included in any of the final Ecosim simulations.

Once the optimal vulnerability search was complete, vulnerability values (v) were inspected to ensure they were ecologically realistic. Although some adjustments to vulnerabilities and feeding time parameters often resulted in better visual fits of the model to observed data, they often resulted in a poorer statistical fit (lower SS) or required unrealistic parameter values, so they were disregarded. Further detailed descriptions of fitting Ecosim models to time-series data can be found in Christensen *et al.* (2005).

We were also interested in better understanding whether the Ecosim model was improved by including bycatch time series data, since it is common to only have reliable time series for commercially important species, which have generally been subjected to some type of stock assessment. We assessed the influence of bycatch data by removing it from the data used to calibrate the model, and recalibrated the model 10 times, as described above, and selected the

best model with the lowest SS. Our results showed that the mean SS for fits where bycatch was removed was lower ($SS = 52053 \pm SD 1688$) than for fits made using all data ($SS = 55809 \pm SD 1832$). Although removal of bycatch data resulted in a better statistical fit, owing to fewer parameters being estimated, the quality of the model fits to observed data were substantially worse. Figure S2 shows the model fits to observed data using all data and where bycatch data were removed for four target species representing different trophic levels, and two commonly caught shark species. Clearly, the fits to observed data for most species were degraded following bycatch data removal, especially for Large bigeye tuna and Large skipjack. When the bycatch data was removed, the biomass predictions of the two shark bycatch species varied significantly in magnitude and trend.

Table S5. Description of the sources and preparation of time series data used for relevant functional groups to calibrate the Ecosim model of the Warm Pool.

Functional group	Data type	Age classes	Source	Data included in the Ecosim model
Small swordfish	Biomass	1-2 years	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	Biomass estimates for 1952 to 2010 derived from the 2013 MFCL assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) for age classes 1-2 years in Area 1. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	1-2 years	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	The 2013 MFCL assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) provides an estimate for fishing mortality (<i>F</i>) for age classes 1-2 years in Area 1 for each year of the time series from 1952 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	1-2 years	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Large swordfish	Biomass	3-20 years	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	Biomass estimates for 1952 to 2010 derived from the 2013 MFCL assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) for age classes 3-20 years in Area 1. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	3-20 years	Davies <i>et al.</i> (2013), Simon Hoyle (SPC)	The 2013 MFCL assessment for southwest Pacific swordfish (Davies <i>et al.</i> 2013) provides an estimate for fishing mortality (<i>F</i>) for age classes 3-20 years in Area 1 for each year of the time series from 1952 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	3-20 years	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Blue marlin	Biomass	All age classes	Kleiber <i>et al.</i> (2003), Simon Hoyle (SPC)	Biomass estimates for 1955 to 1996 derived from the SW area combination of two blocks in a MFCL Pacific-wide stock assessment for blue marlin (Kleiber <i>et al.</i> 2003) for all age classes. Biomass reduced to an index by dividing each year of the time series by the last value in the time series (1996). Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	All age classes	Kleiber <i>et al.</i> (2003), Simon Hoyle (SPC)	The MFCL Pacific-wide stock assessment for blue marlin (Kleiber <i>et al.</i> 2003) provides an estimate for fishing mortality (<i>F</i>) for all age classes in the SW area combination of two blocks for each year of the time series from 1955 to 1996. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	All age classes	Unpublished SPC scientific observer	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Annual estimates were divided

			data (T. Lawson and V. Allain - SPC)	by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Striped marlin	Biomass	All age classes	Davies <i>et al.</i> (2012), Simon Hoyle (SPC)	Biomass estimates for 1952 to 2010 derived from the South Pacific area of the MFCL stock assessment for the southwestern Pacific Ocean (Davies <i>et al.</i> 2012), combined for age classes. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	All age classes	Davies <i>et al.</i> (2012), Simon Hoyle (SPC)	The South Pacific area of the MFCL stock assessment for the southwestern Pacific Ocean (Davies <i>et al.</i> 2012) provides an estimate for fishing mortality (<i>F</i>) for all age classes for each year from 1952 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Other billfish	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Mako shark	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Blue shark	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Other sharks	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Silky shark	Biomass	All age classes	Rice and Harley (2013)	Biomass estimates for 1995 to 2010 derived from the WCPO area bounded by 30°N-30°S, a combination of 2 blocks in a Stock Synthesis stock assessment model for silky shark (Rice and Harley 2013), combined for all age classes. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	All age classes	Rice and Harley (2013)	The Stock Synthesis stock assessment model for silky shark in the WCPO (Rice and Harley 2013) provides an estimate for fishing mortality (<i>F</i>) for all age classes for each year from 1995 to 2010 in a combination of 2 blocks bounded by 30°N-30°S. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.

	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
White tip shark	Biomass	All age classes	Rice and Harley (2012)	Biomass estimates for 1995 to 2010 derived from the WCPO area bounded by 30°N-30°S, a combination of 2 blocks in a Stock Synthesis stock assessment model for White tip shark (Rice and Harley, 2012), combined for all age classes. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	All age classes	Rice and Harley (2012)	The Stock Synthesis stock assessment model for White tip shark (Rice and Harley, 2012) provides an estimate for fishing mortality (<i>F</i>) for all age classes for each year from 1995 to 2010 in the WCPO area, a combination of 2 blocks bounded by 30°N-30°S. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Small bigeye tuna	Biomass	1-16 quarters	Davies <i>et al.</i> (2011), Simon Hoyle (SPC)	Biomass estimates for 1952 to 2010 derived from the 2011 MFCL western Pacific stock assessment for bigeye tuna (Davies <i>et al.</i> 2011) for age classes 1-16 quarters in Region 3. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	1-16 quarters	Davies <i>et al.</i> (2011), Simon Hoyle (SPC)	The 2011 MFCL western Pacific stock assessment for bigeye tuna (Davies <i>et al.</i> 2011) provides an estimate for fishing mortality (<i>F</i>) for age classes 1-16 quarters in Region 3 for each year of the time series from 1952 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	1-16 quarters	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Based on length frequency examination it was assumed that all BET caught in the pole-and-line fishery were small BET. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Large bigeye tuna	Biomass	17-40 quarters	Davies <i>et al.</i> (2011), Simon Hoyle (SPC)	Biomass estimates for 1952 to 2010 derived from the 2011 MFCL western Pacific stock assessment for bigeye tuna (Davies <i>et al.</i> 2011) for age classes 17-40 quarters in Region 3. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	17-40 quarters	Davies <i>et al.</i> (2011),	The 2011 MFCL western Pacific stock assessment for bigeye tuna (Davies <i>et al.</i> 2011) provides an estimate for fishing mortality (<i>F</i>) for age classes 17-40 quarters in Region 3 for each year of the time

			Simon Hoyle (SPC)	series from 1952 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	17-40 quarters	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The 2005 estimate was divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Small yellowfin tuna	Biomass	1-9 quarters	Langley <i>et al.</i> (2011), Simon Hoyle (SPC)	Annual biomass estimates for 1952 to 2010 derived from the 2011 MFCL western Pacific stock assessment for yellowfin tuna (Langley <i>et al.</i> 2011) for age classes 1-9 quarters in Region 3. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	1-9 quarters	Langley <i>et al.</i> (2011), Simon Hoyle (SPC)	The 2011 MFCL western Pacific stock assessment for yellowfin tuna (Langley <i>et al.</i> 2011) provides an estimate for fishing mortality (<i>F</i>) for age classes 1-9 quarters in Region 3 for each year of the time series from 1952 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	1-9 quarters	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Based on length frequency examination it was assumed that all YFT caught in the pole-and-line fishery were small YFT. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Large yellowfin tuna	Biomass	10-28 quarters	Langley <i>et al.</i> (2011), Simon Hoyle (SPC)	Annual biomass estimates for 1952 to 2010 derived from the 2011 MFCL western Pacific stock assessment for yellowfin tuna (Langley <i>et al.</i> 2011) for age classes 10-28 quarters in Region 3. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	10-28 quarters	Langley <i>et al.</i> (2011), Simon Hoyle (SPC)	The 2011 MFCL western Pacific stock assessment for yellowfin tuna (Langley <i>et al.</i> 2011) provides an estimate for fishing mortality (<i>F</i>) for age classes 10-28 quarters in Region 3 for each year of the time series from 1952 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	10-28 quarters	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline discard biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Juvenile skipjack	Biomass	1 quarter	Hoyle <i>et al.</i> (2011)	Annual biomass estimates for 1972 to 2010 derived from the 2011 MFCL south Pacific stock assessment for skipjack (Hoyle <i>et al.</i> 2011) for age class 1 quarter in Region 2. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	1 quarter	Hoyle <i>et al.</i> (2011)	The 2011? South Pacific stock assessment for skipjack (Hoyle <i>et al.</i> 2011) provides an estimate for fishing mortality (<i>F</i>) for age class 1 quarter in Region 2 for each year of the time series from 1972 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	1 quarter	Unpublished SPC scientific observer	Catch estimates for juvenile skipjack (age class 1 only) in pole-and-line, PSA and PSU in 2005 was taken from Region 2 in the 2011 MFCL stock assessment (Hoyle <i>et al.</i> 2011). Annual estimates divided

			data (T. Lawson and V. Allain - SPC)	by the water surface area of Region 2 of 25,093,000 km ² . Data series assigned a data type code of 6 (catches) in Ecopath.
Small skipjack	Biomass	2-3 quarters	Hoyle <i>et al.</i> (2011)	Annual biomass estimates for 1972 to 2010 derived from the 2011 MFCL south Pacific stock assessment for skipjack (Hoyle <i>et al.</i> 2011) for age classes 2-3 quarters in Region 2. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	2-3 quarters	Hoyle <i>et al.</i> (2011)	The 2011 South Pacific stock assessment for skipjack (Hoyle <i>et al.</i> 2011) provides an estimate for fishing mortality (<i>F</i>) for age classes 2-3 quarters in Region 2 for each year of the time series from 1972 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	2-3 quarters	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Pole-and-line weight estimates were based on observer data in the Warm Pool area and raised to the total pole-and-line fishing effort (in number of days) in the area. Based on length frequency examination it was assumed that 5% of the SKJ caught in the pole-and-line fishery were Small SKJ. Longline weights were estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The annual estimates for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Large skipjack	Biomass	4-16 quarters	Hoyle <i>et al.</i> (2011)	Annual biomass estimates for 1972 to 2010 derived from the 2011 MFCL south Pacific stock assessment for skipjack (Hoyle <i>et al.</i> 201xx) for age classes 4-16 quarters in Region 2. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	4-16 quarters	Hoyle <i>et al.</i> (2011)	The 2011 South Pacific stock assessment for skipjack (Hoyle <i>et al.</i> 2011) provides an estimate for fishing mortality (<i>F</i>) for age classes 4-16 quarters in Region 2 for each year of the time series from 1972 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	4-16 quarters	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Pole-and-line weight estimates were based on observer data in the Warm Pool area and raised to the total pole-and-line fishing effort (in number of days) in the area. Based on length frequency examination it was assumed that 5% of the SKJ caught in the pole-and-line fishery were Large SKJ. Longline weights were estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine catch biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The annual estimates for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Albacore	Biomass	All age classes	Hoyle <i>et al.</i> (2012)	Annual biomass estimates for 1960 to 2010 derived from the 2012 MFCL South Pacific stock assessment for albacore (Hoyle <i>et al.</i> 2012) for all age classes combined in the south Pacific area. Biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
	Fishing mortality	All age classes	Hoyle <i>et al.</i> (2012)	The 2012 South Pacific stock assessment for albacore (Hoyle <i>et al.</i> 2012) provides an estimate of average fishing mortality (<i>F</i>) for all ages combined in the south Pacific area for each year of the time series from 1972 to 2010. Data series assigned a data type code of 4 (fishing mortality forcing) in Ecopath.
	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The annual estimates for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.

Wahoo	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. The annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Dolphinfish	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The annual estimates for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Small tunas	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The annual estimates for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Escolar & oilfish	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Lancetfish	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Pomfret	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. The annual estimates were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Rainbow runner	Landings	All age classes	Unpublished SPC scientific observer data (T. Lawson and V. Allain - SPC)	Longline catch biomass was estimated from models fitted with longline observer data in the WCPO during the period 1994-2008 using splines with 4 degrees of freedom. Purse-seine discard biomass was estimated from models fitted with purse seine observer data in the WCPO during the period 1995-2005. Pole-and-line weight estimates were based on observer data in the Warm Pool area raised to the total pole-and-line fishing effort (in number of days) in the area. Observed fate code was used to distinguish landings from discards. The annual estimates for each fishery were divided by the water surface area of the entire Warm Pool (12,086,900 km ²). Data series assigned a data type code of 6 (catches) in Ecopath.
Epipelagic crustaceans	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass

				reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Epipelagic fish	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Epipelagic small fish	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Epipelagic molluscs	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Migratory mesopelagic fish & crustaceans	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Migratory mesopelagic molluscs	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Mesopelagic fish & crustaceans	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Mesopelagic molluscs	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Highly migratory bathypelagic forage	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Migratory bathypelagic forage	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.

				reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Bathypelagic forage	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Mesozooplankton	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Microzooplankton	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Large phytoplankton	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.
Small phytoplankton	Biomass	All age classes	Lehodey (2010)	Annual biomass estimates for 1997 to 2008 derived from the SEAPODYM model covering the Warm Pool region for albacore as the predator (Lehodey <i>et al.</i> 2008) by dividing the model's estimated forage biomass by the percentage contribution of the group as estimated by field surveys. The group's biomass reduced to an index by dividing each year of the time series by the 2005 value. Data series assigned a data type code of 0 (relative biomass) in Ecopath.

Table S6. Vulnerability (v) values used for Ecosim scenario modelling the Warm Pool model, which were derived after fitting to 110 data time series.

Group no.	Prey	Predator																							
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	Seabirds	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
2	Cetaceans	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
3	Turtles	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
4	Small swordfish	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
5	Large swordfish	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
6	Blue marlin	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
7	Striped marlin	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
8	Billfish other	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
9	Mako shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
10	Blue shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
11	Silky shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
12	White tip shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
13	Other sharks	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
14	Small BET	2.000	2.000	2.000	4.000	2.500	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	2.000	1.700	2.000	5.000	2.000	2.000	2.000	2.000	2.000	1.394	2.000
15	Large BET	2.000	2.000	2.000	2.000	2.500	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
16	Small YFT	2.000	2.000	2.000	2.000	2.500	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	2.000	1.700	2.000	5.000	2.000	2.000	2.000	2.000	2.000	1.394	2.000
17	Large YFT	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
18	Baby SKJ	2.000	2.000	2.000	2.000	2.000	2.000	2.800	2.000	2.000	2.000	2.000	2.000	2.000	1.006	1.700	2.400	5.000	2.000	2.000	3.600	2.000	2.000	1.394	2.000
19	Small SKJ	2.000	2.000	2.000	2.000	2.000	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	2.000	1.700	2.400	5.000	2.000	2.000	2.000	2.000	2.000	1.394	2.000
20	Large SKJ	2.000	2.000	2.000	2.000	2.000	2.000	2.800	2.000	2.000	2.000	2.300	2.345	2.400	2.000	1.700	2.000	5.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
21	Albacore	2.000	2.000	2.000	4.000	2.500	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	2.000	1.700	2.000	5.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
22	Wahoo	2.000	2.000	2.000	2.000	2.000	2.000	2.800	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
23	Dolphinfish	2.000	2.000	2.000	2.000	2.000	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	2.000	1.700	2.400	5.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
24	Small tunas	2.000	2.000	2.000	2.000	2.000	2.000	2.800	2.000	2.000	2.000	2.300	2.345	2.400	2.000	1.700	2.400	5.000	2.000	2.000	3.600	1.950	2.000	1.394	2.000
25	Escolar & Oilfish	2.000	2.000	2.000	4.000	2.500	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	1.700	2.400	5.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
26	Lancetfish	2.000	2.000	2.000	4.000	2.500	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	2.000	2.000	2.000	5.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
27	Opah	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.300	2.345	2.400	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
28	Pomfret	2.000	2.000	2.000	4.000	2.500	2.000	2.800	2.000	2.000	1.600	2.000	2.345	2.400	1.006	1.700	2.400	5.000	2.000	2.000	3.600	1.950	2.000	2.000	2.000
29	Rainbow runner	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.600	2.300	2.345	2.400	2.000	2.000	2.400	5.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
30	Epi crust	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.600	2.000	2.000	2.000	1.006	1.700	2.400	5.000	1.001	2.000	3.600	1.950	2.000	1.394	1.331
31	Epi fish	2.000	2.000	2.000	2.000	2.000	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	2.000	2.000	2.400	5.000	1.001	2.000	3.600	1.950	2.000	1.394	1.331
32	Epi small fish	2.000	2.000	2.000	4.000	2.000	2.000	2.800	2.000	2.000	1.600	2.300	2.000	2.400	1.006	1.700	2.400	5.000	1.001	2.000	3.600	1.950	2.000	1.394	1.331
33	Epi mollusc	2.000	2.000	2.000	4.000	2.500	2.000	2.800	2.000	2.000	1.600	2.300	2.345	2.400	1.006	1.700	2.400	5.000	1.001	2.000	3.600	1.950	2.000	1.394	1.331
34	M Meso fish+other	2.000	2.000	2.000	4.000	2.500	2.000	2.000	2.000	2.000	1.600	2.000	2.000	2.000	1.006	1.700	2.400	5.000	2.000	2.000	3.600	1.950	2.000	1.394	1.331
35	M meso mollusc	2.000	2.000	2.000	4.000	2.500	2.000	2.000	2.000	2.000	1.600	2.000	2.000	2.000	1.006	1.700	2.400	5.000	2.000	2.000	3.600	1.950	2.000	1.394	1.331
36	Meso fish + other	2.000	2.000	2.000	4.000	2.500	2.000	2.800	2.000	2.000	2.000	2.000	2.000	2.400	1.006	1.700	2.400	5.000	2.000	2.000	3.600	1.950	2.000	1.394	1.331
37	Meso mollusc	2.000	2.000	2.000	2.000	2.500	2.000	2.800	2.000	2.000	2.000	2.000	2.000	2.400	1.006	1.700	2.400	5.000	2.000	2.000	3.600	1.950	2.000	2.000	2.000
38	HM Bathy forage	2.000	2.000	2.000	4.000	2.500	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.006	1.700	2.400	5.000	2.000	2.000	3.600	1.950	2.000	1.394	1.331
39	M Bathy forage	2.000	2.000	2.000	4.000	2.500	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.006	1.700	2								

Table S6 continued.

Group no.	Prey	Predator																	
		25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
1	Seabirds	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
2	Cetaceans	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
3	Turtles	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
4	Small swordfish	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
5	Large swordfish	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
6	Blue marlin	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
7	Striped marlin	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
8	Billfish other	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
9	Mako shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
10	Blue shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
11	Silky shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
12	White tip shark	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
13	Other sharks	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
14	Small BET	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
15	Large BET	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
16	Small YFT	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
17	Large YFT	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
18	Baby SKJ	2.000	2.000	2.000	2.000	1.103	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
19	Small SKJ	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
20	Large SKJ	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
21	Albacore	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
22	Wahoo	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
23	Dolphinfish	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
24	Small tunas	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
25	Escolar & Oilfish	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
26	Lancetfish	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
27	Opah	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
28	Pomfret	2.000	20.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
29	Rainbow runner	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
30	Epi crust	2.000	20.000	2.000	1.016	1.103	2.000	1.562	1.178	3.182	1.933	5.000	2.000	2.000	2.116	2.000	2.000	2.000	2.000
31	Epi fish	2.000	20.000	2.000	2.000	1.103	2.000	2.000	2.000	3.182	1.933	5.000	2.000	2.000	2.116	2.000	2.000	2.000	2.000
32	Epi small fish	2.000	20.000	2.000	2.000	1.103	2.000	1.562	2.000	3.182	1.933	5.000	2.000	2.000	2.116	2.000	2.000	2.000	2.000
33	Epi mollusc	2.000	20.000	2.000	1.016	1.103	2.000	1.562	1.178	2.000	1.933	5.000	2.000	2.000	2.116	2.000	2.000	2.000	2.000
34	M Meso fish+other	2.000	20.000	2.000	1.016	1.103	2.000	2.000	2.000	3.182	2.000	5.000	1.738	5.500	2.116	1.612	2.000	2.000	2.000
35	M meso mollusc	2.000	20.000	2.000	2.000	1.103	2.000	2.000	2.000	3.182	2.000	2.000	1.738	5.500	2.116	1.612	2.000	2.000	2.000
36	Meso fish + other	2.000	20.000	2.000	1.016	1.103	2.000	2.000	2.000	2.000	2.000	2.000	2.000	5.500	2.000	1.612	2.000	2.000	2.000
37	Meso mollusc	2.000	20.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.738	2.000	2.000	1.612	2.000	2.000	2.000
38	HM Bathy forage	2.000	20.000	2.000	1.016	1.103	2.000	2.000	2.000	3.182	2.000	2.000	1.738	5.500	2.000	1.612	2.000	2.000	2.000
39	M Bathy forage	2.000	20.000	2.000	1.016	2.000	2.000	2.000	2.000	3.182	2.000	2.000	1.738	5.500	2.000	2.000	2.000	2.000	2.000
40	Bathy forage	2.000	20.000	2.000	1.016	2.000	2.000	2.000	2.000	3.182	2.000	2.000	1.738	2.000	2.000	1.612	2.000	2.000	2.000
41	Mesozpk	2.000	20.000	2.000	2.000	1.103	2.000	1.562	1.178	3.182	1.933	5.000	1.738	5.500	2.116	1.612	2.000	2.000	2.000
42	Microzpk	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.178	3.182	1.933	2.000	1.738	2.000	2.000	2.000	2.000	2.000	2.000
43	Large phyto	2.000	2.000	2.000	2.000	2.000	2.000	1.562	1.178	2.000	1.933	2.000	1.738	2.000	2.116	1.612	2.000	2.000	2.000
44	Small phyto	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.178	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
45	Detritus	2.000	2.000	2.000	2.000	2.000	2.000	1.562	1.178	3.182	1.933	5.000	1.738	5.500	2.116	1.612	2.000	2.000	2.000
46	Fishery discards	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	1.612	2.000	2.000	2.000

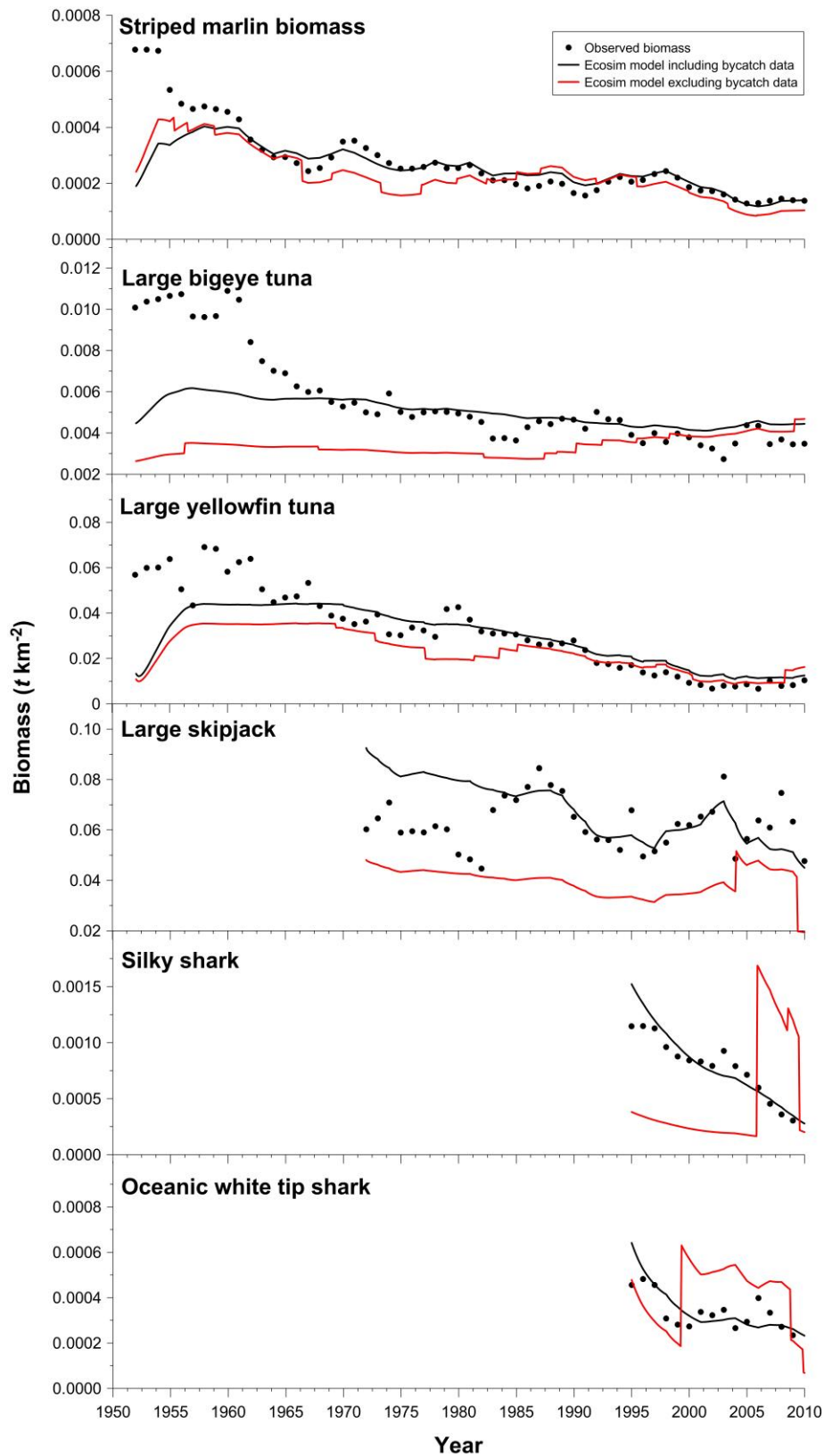


Figure S2. Observed (dots) and predicted (lines) biomass ($t\ km^{-2}$) for four target species (*Striped marlin*, *Large bigeye tuna*, *Large yellowfin tuna*, *Large skipjack*) and two bycatch species (*Silky* and *Oceanic white tip sharks*) when calibrating Ecosim models using time series for all species (black lines) and where bycatch data were excluded (red lines). Note the poor biomass predictions for *Silky* and *Oceanic white tip sharks* once their data were excluded from the calibration.

Appendix S4 – Ecological indicators

Mixed Trophic Impact

Mixed Trophic Impact (MTI) is a sensitivity analysis within Ecopath that allows the strongest trophic interactions to be identified. This can be useful both in terms of understanding important trophic relationships, but also for identifying any possible errors or anomalies that may have occurred during model construction or the model balancing process. Results of MTI are best represented graphically—owing to the large number of combinations of predator-prey interactions—and are shown in Figure 2 in the main manuscript.

Keystone groups

The MTI analysis primarily identifies the strongest trophic interactions as a result of abundant and/or productive species preying upon species with lower biomass and productivity. However, some of the most important ecological interactions can occur when species that have relatively low biomass have an important predatory role in structuring an ecosystem. These species are often considered “keystone” species and can have a disproportionately large influence on the structure of the ecosystem relative to their biomass (Paine 1995). Results from Ecopath’s “Keystoneness” analysis for the Warm Pool model are given in Table S7, showing Ecopath’s keystone index for each functional group and its relative total impact on the ecosystem.

Table S7. Results from Ecopath’s “Keystoneness” analysis of the Warm Pool model, showing the keystone index and the relative total impact measure for each functional group.

Group No.	Functional group name	Keystone index	Relative total impact
1	Seabirds	-0.957	0.084
2	Cetaceans	-2.193	0.005
3	Turtles	-1.981	0.008
4	Small swordfish	-1.125	0.057
5	Large swordfish	-0.659	0.166
6	Blue marlin	-0.711	0.148
7	Striped marlin	-1.183	0.050
8	Other billfish	-0.888	0.098
9	Mako shark	-1.188	0.049
10	Blue shark	-1.394	0.031
11	Silky shark	-0.520	0.229
12	Oceanic whitetip shark	-1.123	0.057
13	Other sharks	-0.180	0.501
14	Small bigeye tuna	-1.021	0.072
15	Large bigeye tuna	-0.756	0.133
16	Small yellowfin tuna	0.030	0.814
17	Large yellowfin tuna	-0.208	0.470
18	Juvenile skipjack	-0.847	0.108
19	Small skipjack	-0.534	0.222
20	Large skipjack	-0.213	0.466
21	Albacore	-0.158	0.528
22	Wahoo	-1.501	0.024
23	Dolphinfish	-0.607	0.188
24	Small tunas	-0.758	0.133
25	Escolar & Oilfish	-0.808	0.118
26	Lancetfish	-0.465	0.260
27	Opah	-1.078	0.063
28	Pomfret	-0.686	0.156
29	Rainbow runner	-0.824	0.114
30	Epipelagic crustaceans	-0.451	0.363
31	Epipelagic fish	-0.160	0.558
32	Epipelagic small fish	-0.028	0.830
33	Epipelagic molluscs	0.034	0.870
34	Migratory mesopelagic fish & crustaceans	-0.278	0.412
35	Migratory mesopelagic molluscs	-0.341	0.350
36	Mesopelagic fish & crustaceans	0.117	1.000
37	Mesopelagic molluscs	-0.058	0.667
38	Highly migratory bathypelagic forage	-0.448	0.275
39	Migratory bathypelagic forage	-0.441	0.276
40	Bathypelagic forage	-0.363	0.330
41	Mesozooplankton	-0.212	0.509
42	Microzooplankton	-0.249	0.462
43	Large phytoplankton	-0.734	0.145
44	Small phytoplankton	-0.441	0.346
45	Detritus	-	-
46	Fishery discards	-	-

Ecological indicators

Fulton *et al.* (2005) undertook a simulation study using two ecosystem models and found that indicators at the community or ecosystem level were the most reliable indicators of ecosystem change under fishing. They found no single indicator was able to provide a complete representation of the ecosystem and suggest that simultaneous use of a variety of indicators is likely to be required to detect the full range of impacts from fishing.

Fulton *et al.* (2005) found that useful ecosystem indicators from Ecopath are the Fishing in Balance index (FIB) and the average trophic level of the catch (TL_c). In a recent Ecopath modelling study investigating the effects of fisheries and climate change on the northwest Pacific marine ecosystem, Ainsworth *et al.* (2011) found that Kempton's Q index adapted for ecosystem models (see Ainsworth and Pitcher 2006) was a particularly useful indicator for assessing changes in the diversity and biomass of high trophic level species ($>TL\ 3$). Such an indicator is highly relevant to the Warm Pool ecosystem since the pelagic fisheries in the region target large predatory tunas, billfish and tuna-like species. Therefore, FIB, TL_c and Kempton's Q were three indicators of ecosystem change chosen for use in scenario modelling of the Warm Pool model. A description of each indicator is provided below.

Fishing in Balance (FIB) index

The FIB index (Pauly *et al.* 2000) provides an indication whether fisheries are balanced in ecological terms and not causing disruption to the functionality of the ecosystem. FIB is defined for any year i as:

$$FIB = \log \left(Y_i \cdot \left(\frac{1}{TE} \right)^{TL_i} \right) - \log \left(Y_0 \cdot \left(\frac{1}{TE} \right)^{TL_0} \right)$$

where Y is the catch, TL is the mean trophic level of the catch, TE is the transfer efficiency (i.e. the ratio between the sum of export from a specified trophic level plus the flow that is transferred to

the next trophic level, and the throughput of the specified trophic level), while 0 is the year used as a baseline, in this case 2005. FIB can provide an indication of overfishing when catches do not increase as expected (or a decrease in TL_c) given available productivity in the system, or if the effects of fishing are sufficient to compromise the functionality of the ecosystem ($FIB < 0$). In contrast, FIB can indicate an expansion of the fishery (e.g. increase in diversity and/or biomass of bycatch) ($FIB > 0$).

Mean trophic level of the catch (TL_c)

The mean trophic level of the catch (TL_c) is an indicator whether fisheries are changing their fishing or targeting practices in response to changes in the abundance or catchability of traditional target species. For example, declines in the abundance of large predatory fish by overfishing has resulted in fisheries moving to target species at lower trophic levels in order to maintain profitability (Pauly *et al.* 1998a). TL_c is calculated for year y as:

$$TL_{cy} = \frac{\sum_i (TL_i \cdot Y_{iy})}{\sum_i Y_{iy}}$$

Where Y_{iy} is the catch of functional group i , and TL_i is the trophic level of functional group i .

Kempton's Q index

Kempton's Q index describes the slope of the cumulative species log-abundance curve and measures the diversity and evenness of the component of the ecosystem composed of species or functional groups with a trophic level greater than 3 (Kempton and Taylor 1976, Ainsworth and Pitcher 2006). This was considered a particularly useful indicator for assessing ecosystem changes as targets or bycatch in the Warm Pool fisheries generally occupy trophic levels higher than 3. An increase in the index indicates an increase in upper level evenness and richness (Kempton and

Taylor 1976, Shannon *et al.* 2009), whereby Ecopath uses biomass as a proxy of abundance for each group in estimating evenness. The index is defined as:

$$Q = \frac{0.8 \cdot S}{\log \left(\frac{R_2}{R_1} \right)}$$

where S is the total number of functional groups in the model; R_1 and R_2 are the biomass values of the 10th and 90th percentiles in the cumulative abundance distribution.

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